A PROJECT REPORT ON

Head on collision prevention system for vehicles (Locomotives)

SUBMITTED BY

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DECLARATION

We, hereby declare that the discussion entitled **"Head on collision prevention system for vehicles (Locomotives)"** being submitted by us towards the partial fulfilment of the degree of Bachelor of Technology (B. Tech), in the Department of Electrical Engineering is a project work carried by us, under the supervision of **Mr. Suman Ghosh** and has not been submitted anywhere else.

We will be solely responsible if any kind of plagiarism is found.

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To whom it may Concern

This is to certify that the project work entitled **Head on collision prevention system for vehicles (Locomotives)** is the bona-fide work carried out by the students of Bachelor of Technology in the Department of Electrical Engineering, Techno Main Salt Lake affiliated to Maulana Abul Kalam Azad University of Technology (MAKAUT), West Bengal, India, during academic year 2023-2024, in partial fulfillment of the requirements for the degree of Bachelor of Technology in Electrical Engineering and that this project has not been submitted previously for the award of any other degree, diploma and fellowship.

Suman Ghosh. 06/06/2024

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ABSTRACT

Head-on collisions between trains on the same track are a significant safety risk in Indian railway transportation. Incidents like the 1995 Firozabad rail disaster, which resulted in 358 deaths, and the 1999 Gaisal train collision, which caused 285 fatalities, highlight the urgent need for better safety systems. Despite advancements, Indian Railways continue to face challenges in ensuring the safety of millions of passengers and vast quantities of goods transported daily. Factors such as high traffic density, human errors, and outdated signaling systems increase the risk of accidents. To address these challenges, we present the "**Head-on Collision Prevention System for Locomotives**." This system uses advanced sensors and braking technology to detect obstacles, monitor track conditions, and stop trains in time to prevent accidents.

Our model uses DC motors connected to L298N motor drivers to move the trains, powered by Arduino and Li-ion batteries. A Sharp-IR sensor detects obstacles or other trains within 80 cm and signals the motor driver to start braking, slowing the train gradually.

Additionally, the system includes two ultrasonic sensors placed 0.6 meters apart on the track to detect the presence and speed of trains. These sensors measure the speed by calculating the time it takes for a train to pass between them, with a response time of less than 100 milliseconds. An LED indicator warns if another train is on the same track. The system also uses an MPU6050 accelerometer and gyroscope to detect unusual movements, like crashes, and triggers an alert through a buzzer or LED. In case of an accident, the system sends an immediate alert to the nearest control room for a quick response.

This model combines ultrasonic sensors, IR sensors, and gyroscopic technology to create a reliable safety system. By improving the awareness and response of trains, our system aims to **reduce the risk of head-on collisions, protecting lives, cargo, and railway infrastructure**. Historical data suggests that with such systems, incidents like the Firozabad and Gaisal train disasters could be avoided, highlighting the importance of advanced safety measures in Indian railways.

The need for this project is further underscored by the growing demand for railway services in India. The Indian Railways, one of the world's largest rail networks, carries over 23 million passengers daily and transports more than 1.2 billion tons of freight annually. **With the increasing frequency of trains and the complexity of managing extensive rail operations, the implementation of advanced safety systems is crucial. Our collision prevention system aims to enhance the safety and reliability of Indian Railways, contributing to the protection of human lives, minimizing economic losses from accidents, and ensuring the efficient operation of the railway network.**

CHAPTER 1: INTRODUCTION

1.1 What is Head on Collision?

A head-on collision is also commonly referred to as a **frontal collision**. These types of vehicular crashes usually occur when two trains that are coming in **opposite directions collide into one another**. When a train strikes a stationary object such as a tree, light pole, or cement barrier or a movable object, it could be classified as a head-on collision.

Head-on collisions can result in catastrophic **accidents, leading to loss of life, property damage, and environmental harm.** In our daily life, we hear about train accidents that humans, animals and vehicles damaged with this incident. Train vs Train collision is one big accident in all of them when two train come in same track and no other way to escape they collide, and many people died and injured.

1.2 Historical Overview of Train Collisions in Indian Railways

As we know, the Indian Railways is **one of the largest railway networks in the world covering 120,000 km (approx.) of track, over a route of 68000 km (approx.) and over 7000 stations.**

Figure 1.1: Train collision at Kacheguda railway station in Hyderabad, India. The incident occurred on November 11, 2019

On average, at least one person dies in a train accident every 2-3 minutes. Per year, 3-4 millions of people are dangerously injured by these train accidents. These accidents occurred due to many reasons such as the negligence of railway staff like Station Masters, Loco pilots, Track machine workers etc., due to human and equipment failures. [1]

Table 1.1: The following below table shows the Different Collisions occurred between the Trains in the year 2009 to 2014: -

Year	**Track Problems**	**Crossing Accidents**	**Collisions**	**Other Problems**	**Total Accidents**
2009-10	82	71	9	6	168
2010-11	80	54	5	3	142
2011-12	57	62	6	6	134
2012-13	49	59	4	8	122
2013-14	52	51	3	10	117
Total	320	297	27	33	683

Table 1.2: Date with Incidents

Date	Incidents
12 December 1973	Rear end Collision between 358 Up Itarsi-Bhusaval Passenger and Stationary Goods Train No. L-30 Up at Burhanpur Station of Central Railway.

22 November 1984	The Mumbai local rail disaster occurred where one train collided with another train on the Byculla railway station.
14 December 2004	The Jammu Tawi Express collided head-on with the DMU Jalandhar–Amritsar passenger train about 10 km (6.2 mi) from Mukerian town, Hoshiarpur, Punjab, killing 37 passengers and injuring 50. The accident was allegedly caused due to a fault in the telephone line, which prevented proper signal warnings.
26 December 2012	Head on collision between 55908 DN Ledo – Dibrugarh Town passenger and Light Engine between Chabua and Panitola on New Tinsukia – Dibrugarh Town section of Tinsukia division of Northeast Frontier Railway. Injuring 71 persons.
2 June 2023	2023 Odisha train collision: Train 12841 Coromandel Express running at 128 km/h (80 mph) collides with a freight train (goods train) loaded with iron ore in Odisha's Balasore district.

1.3 Reason of Head on Collision

- **High Traffic Density:** Indian Railways has one of the highest traffic densities in the world, leading to congested tracks and increased chances of collisions.
- **Human Error:** Mistakes by train operators, signalmen, or other railway staff can lead to miscommunications and mismanagement of train movements, resulting in head-on collisions.
- **Outdated Signaling Systems:** Many parts of the Indian railway network still rely on old and inefficient signaling systems, which are prone to failure and misinterpretation.
- **Insufficient Communication Infrastructure:** Poor communication infrastructure can lead to delays in transmitting crucial information about train movements and track conditions, increasing the risk of accidents.
- **Track Switching Errors:** Errors in track switching mechanisms can inadvertently place two trains on the same track, leading to potential collisions.
- **Inadequate Safety Protocols:** Lack of stringent safety protocols and regular training for railway staff can contribute to operational lapses and accidents.
- **Weather Conditions:** Adverse weather conditions like fog, heavy rain, or storms can impair visibility and affect signaling systems, leading to increased risk of collisions.
- **Technical Failures:** Mechanical or technical failures in trains or signaling equipment can lead to unplanned stops or misdirections, causing head-on collisions.

1.4 Effects of head on collision

When two trains at very high speeds collide, the impact would be extremely destructive. The kinetic energy of the trains would be converted into other forms of energy, **causing significant damage to the trains and posing a serious risk to the passengers and crew on board.** The force of the impact would depend on various factors such as the speed and mass of the trains, as well as the angle of collision. Train crashes at high speeds can result in catastrophic **consequences, including loss of life, injuries, and extensive damage to the trains and surrounding infrastructure, environmental harm, property damage etc.** Due to head-on collision, there will be a time delay in schedule i.e. trains can't be able to reach stations in time. It often led to investigations to determine the cause of the accident. Insurance claims and compensation for victims and their families can result in significant financial costs for the groups involved.

Due to head-on collision, psychological impact is also there. Post-traumatic stress disorder (PTSD), anxiety, depression, and also other mental health issues may develop as a result of the trauma experienced during and after the accident. [1]

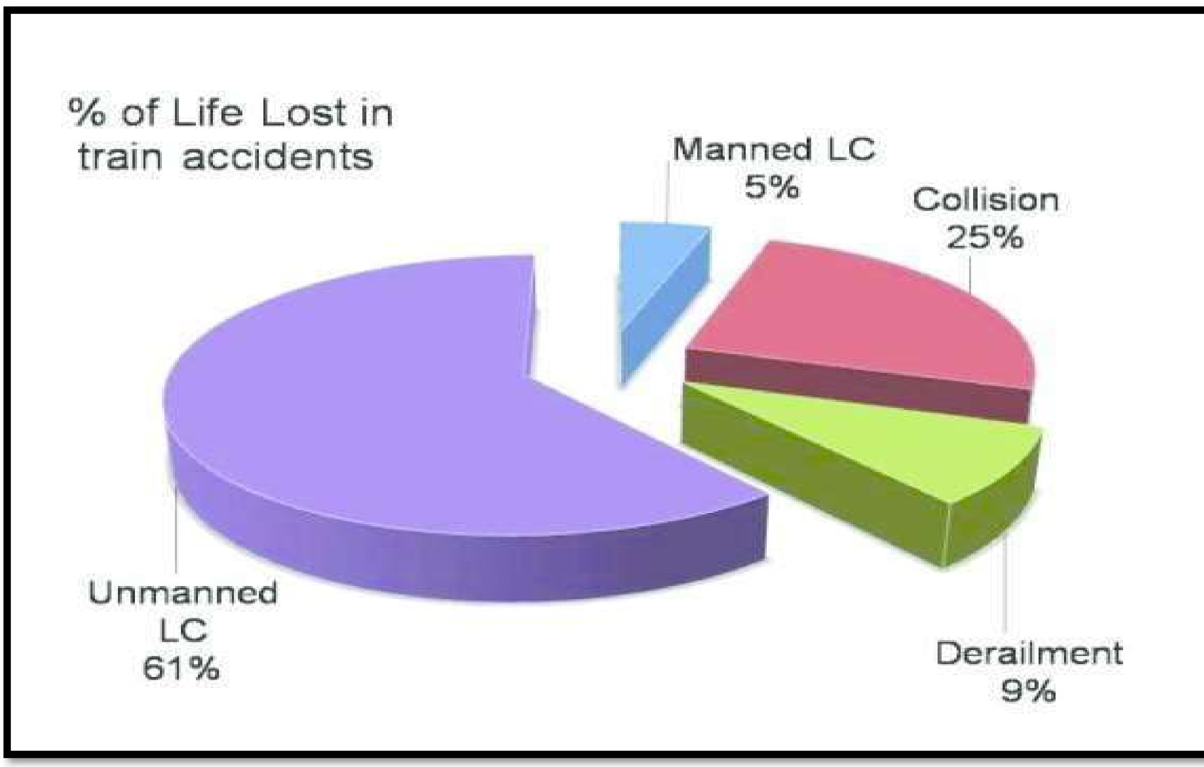


Figure 1.2: A chart showing the loss of lives in train accidents (in percentage)

CHAPTER 2: NEED FOR HEAD ON COLLISION PREVENTION SYSTEM IN LOCOMOTIVES

The necessity of this project is highlighted by the rising demand for railway services in India. As one of the largest rail networks globally, **Indian Railways transports over 23 million passengers daily and handles more than 1.2 billion tons of freight each year.** The increasing frequency of trains and the complexity of managing such a vast network make the adoption of advanced safety systems essential. **Our collision prevention system seeks to improve the safety and reliability of Indian Railways, safeguarding lives, reducing economic losses from accidents, and ensuring the efficient operation of the railway network.**

2.1: Head on collision prevention system for Locomotives

A head-on collision prevention system for trains is a safety technology designed to reduce the risk of head-on collisions between two trains traveling on the same track. Head-on collisions can result in catastrophic accidents, leading to loss of life, property damage, and environmental harm. These systems employ various sensors, communication technologies, and automation to detect and prevent such collisions.

The specific implementation of a head-on collision prevention system may vary depending on the country, railway network, and the technology in use. The primary goal is to ensure the safety of passengers, crew, and cargo by minimizing the risk of head-on collisions and other potential accidents on the railway.

Table 2.1: Head on Collision Accidents dates

Date	Head on Collision Accidents
24 October 1907	A passenger train collided with a freight train at Ko Lakhpat station, killing 11 and injuring 27
27 April 1920	Doon Express train collided with a goods train near Moradabad, resulting in 120 deaths and 50 injuries
June 1953	A passenger train and freight train collided between Sempaore and Kalihar, killing 5 and injuring 50.
23 July 1957	Tatanagar Train collision happened outside the Down outer signal of Tatanagar between Gua-Tata passenger and 304 Down Hazaribagh-Ranchi-Howrah Express running on the Adityapur-Tatanagar section.
8 November 1957	8 November 1957 – Train collision at Kosma Railway Station, on the Shikohabad-Farrukhabad Broad Gauge single line section of the Northern Railway.

4 January 1961	Two passenger trains collided head-on near Umesh Nagar, killing 35 and injuring 61.
14 July 1969	A freight train crashed into a stationary passenger train at Jaipur, killing 85 and injuring 130.
26 October 1970	Head-on Collision between 1 Ajd Up Passenger Train and 456 Down Goods Train at Khurja Jn. Station of Northern Railway
25 April 1971	Head-on Collision between No. 102 Do-4 Up Goods Train and No. 395 Wardha Kazipet Passenger Train Near Asifabad Road Station of the South Central Railway
12 December 1973	Rear end Collision between 358 Up Itarsi-Bhusaval Passenger and Stationary Goods Train No. L-30 Up at Burhanpur Station of Central Railway.
26 October 1979	Rear End Collision between Electric Multiple Unit Trains No. E-65 and No. E-67 on The Down Suburban Line at Madras Egmore Station on Madras Beach Tambaram Suburban Metre Gauge Section of Southern Railway.
22 November 1984	The Mumbai local rail disaster occurred where one train collided with other train on the Byculla railway station.
14 December 2004	The Jammu Tawi Express collided head-on with the DMU Jalandhar–Amritsar passenger train about 10 km (6.2 mi) from Mukerian town, Hoshiarpur, Punjab, killing 37 passengers and injuring 50. The accident was allegedly caused due to a fault in the telephone line, which prevented proper signal warnings.
2 June 2023	2023 Odisha train collision: Train 12841 Coromandel Express running at 128 km/h (80 mph) collides with a freight train (goods train) loaded with iron ore in Odisha's Balasore district.

2.2 : Existing Collision Prevention Systems in Indian Locomotives

There are some existing Collision Prevention System by which it can helps us to prevent the deadly Head-on collision. These are the system indigenously developed by Indian Railways through Research Designs & Standards Organization (RDSO).

2.2.1 : Train Protection and Warning Systems (TPWS)

Its main objective is to make provision of cost-effective Train Protection & Warning System to mitigate the risk of Signal Passing at Danger (SPAD) by Motormen/Loco Pilots of trains leading to accidents. Train Protection and Warning System conforming to Open Specifications, Interoperable, Upgradeable, Multiple Vendor with indigenized manufacturing for long term sustained support.
It assures a higher level of safety during train operation. It allows safe movement of trains under its supervision. It enables automatic train protection and prevents collision, like situation. It facilitates running the train at maximum permitted speed by providing the indication to the driver 500 meters in advance of signal and higher average speed of train. The Entire system provides assistance to the Driver and can be called as an aid to the Driver.

- TPWS basically consists of: (i) Cab Equipment (ii) Trackside Equipment
- Cab Equipment/On-Board Equipment

It consists of the following units: (i) On-Board Computer (OBC) (ii) Balise Transmission Module (BTM) (iii) BTM Antenna (iv) Simplified Driver Machine Interface (SDMI) (v) Wheel Sensors (vi) Brake Interface (vii) Health indication Panel cum Emergency Brake (EB) counter (viii) Audio Buzzer unit (ix) Isolation switch

- Trackside Equipment

It consists of the following: (i) Line Side Electronic unit (LEU) (ii) Euro Balise consists of
□ Fixed Balise □ Controlled Balise. **[2]**

Figure 2.1: Train Protection and Warning Systems (TPWS)

2.2.2 : Automatic Train Protection (ATP) Systems

Automatic Train Protection (ATP) system is a control system used by railways to help avoid collisions by automatically controlling the maximum permissible speed (MPS) that a train can travel at in at any given time relative to its movement authority.
The primary goal of ATP is to prevent **accidents caused by human error**, signal violations or other unforeseen circumstances. ATP systems utilize a combination of sensors, communication devices, and on-board control systems to monitor and manage train speeds, ensuring compliance with predetermined safety parameters.

The various factors determining the throughput are:
• Running Speed. • Block section length. • Block overlap. • Train length. • Station section. • Sighting distance. • Running time in block section • Time taken for the train length to clear block overlap

The system constantly monitors the position and speed of trains and intervenes if it detects a potential collision risk. By automatically controlling train speeds and maintaining safe separation distances, ATP significantly reduces the likelihood of accidents caused by human error or miscommunication.

ATP systems continuously monitor and enforce speed limits for trains based on factors such as track conditions, signal status, and the presence of other trains. This prevents trains from exceeding safe speeds, mitigating the risk of derailments or other accidents associated with high-speed travel.

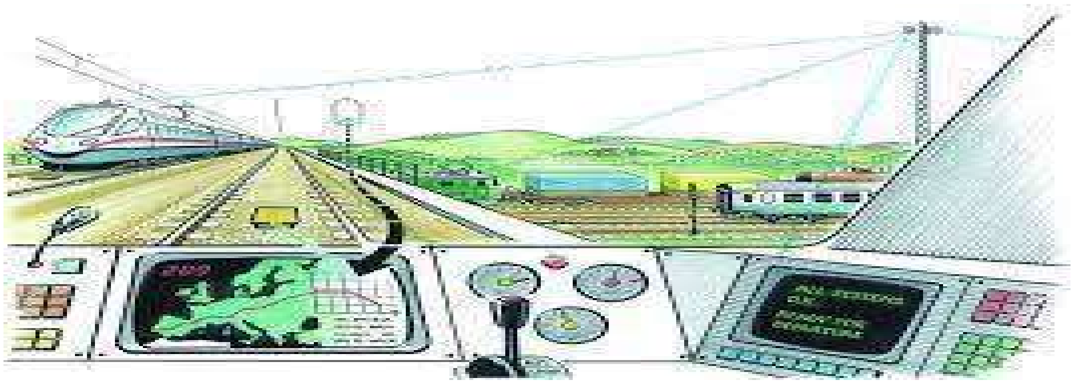


Figure 2.2: Automatic Train Protection (ATP) Systems

2.2.3 : Indian Train Collision Avoidance System (ITCAS)

Indian Train Collision Avoidance System (ITCAS) is an indigenously developed Automatic Train Protection (ATP) System meant to provide protection to trains against Signal Passing at Danger (SPAD), excessive speed and collisions. **ITCAS provides continuous update of Movement Authority (distance up to which the train is permitted to travel without danger).** Hence during unsafe situations when brake application is necessitated, and the Crew has either failed to do so, or is not in position to do so, automatic brake application shall take place. TCAS has additional features to display information like speed, location, distance to signal ahead, Signal aspects etc. in Loc Pilot's cab and generation of Auto and Manual SOS messages (Distress messages) from Loco as well as Station unit in case of emergency situation.**[2]**

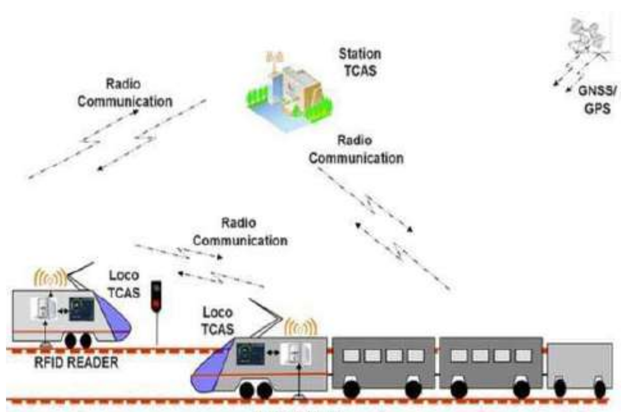


Figure 2.3: Indian Train Collision Avoidance System (ITCAS)

2.3 : Technologies Employed in Indian Locomotive Safety

Technology can play a vital role in improving railway safety in India by developing and implementing systems that can prevent collisions, detect faults, monitor tracks, and enhance communication. According to a report by the Indian Railway, the **Technology Mission on Railway Safety** was launched in 2003 in collaboration with the Department of Science and Technology, IIT Kanpur and the Consortium of Industries to achieve higher level of safety in train operation. Some of the technologies developed under this mission include **anti-train collision system (Kavach), wheel impact load detector, ultrasonic broken rail detection system, and train protection and warning system.**

2.3.1 Indian Railways Signal and Telecommunication Infrastructure

Trains carry more loads resulting in higher traffic capacity since trains move on specific tracks called rails, their path is to be fully guided and there is no arrangement of steering. Clear of obstruction as available with road transportation, so there is a need to provide control on the movement of trains in the form of Railway signals which indicate the drivers to stop or move and also the speed at which they can pass a signal.

Since the load carried by the trains and the speed which the trains can attain are high, they need more braking distance before coming to the stop from full speed. Without Signal to be available on the route to constantly guide the driver accidents will take place due to collisions.

There are basically two purposes achieved by railway signaling.
1. To safely receive and dispatch trains at a station.
2. To control the movements of trains from one station to another after ensuring that the track on which this train will move to reach the next station is free from movement of another train either in the same or opposite direction. This Control is called block working. Preventing the movement from opposite direction is necessary in single line track as movements in both directions will be on the same track.

Figure 2.4: **Indian Railways Signal and Telecommunication Infrastructure**

Optical Fiber Cable (OFC) is laid along the track to provide reliable and noise free communication. OFC network is widely used for Railway Control Communication taking advantage of its all-long haul high bandwidth circuit interconnecting Railway Telephone Exchange. The Passenger Reservation System, Unreserved Ticketing System, Network Freight Operating Management system have been transferred through railway OFC. In **Railway distribution of various media for Telecommunication is as follows:**

1. OFC and RE quad cable in Electrified sections
2. OFC and 4/6 quad cable
3. Only OFC.
4. Railway owned overhead line.
5. Rented overhead line/ channels/ bandwidth from BSNL

In Railway telecom network is supported by Railway Owned MW network using state of the art technology (Digital Microwave System). MW network is spread over Chennai- Jolarpettai, erode – Palghat, Chennai - Tiruchchirappalli, Madurai - Palghat covering all divisional headquarters, mostly along the tracks. There is an in-house Railway Telephone Network connecting all-important offices, officials, Way stations, Divisional Headquarters & Zonal Head Quarters. Railway telephones exchanges are inter-connected through Railway OFC network, Railway Microwave network and are supported by rented BSNL channels as standby. **[3]**

2.3.2 Indigenous Innovations in Train Control and Management Systems

By the term Innovation, we mean a long list of technological breakthroughs that have been happening to the Railways in recent years. Among other, there are zero emission hydrogen trains, advanced autonomous trains able to handle even emergency situations, drone-based automation of infrastructure maintenance, implementation of **4G technology in train management systems, implementation of big data technology** for improving the safety, security and reliability of rail infrastructure, artificial intelligence used for real-time traffic simulations and delay forecasting, variable railway bogies allowing a single train to operate at two or even more different track gauges and many other.
Rail Technology continues to develop as rail customers are demanding faster, safer and more reliable service, while on the other, the railway system is, by definition, suitable for automation of certain processes due to fact that the freedom movement of trains is well taken away by the rails.
Some of the innovations done were:

- In-house development and production of PPEs, Coveralls, Hand Sanitizers, Masks, material delivery robots at hospitals etc, during Lockdown. Efforts were made to design the Ventilators.
- Running of Time Tabled Parcel Special Trains & freight trains and Kisan Rail.
- Digital Initiatives in Passenger Segment as Mobile Ticketing, QR-codes in tickets for Contactless Ticketing and Rail Travel, Chhalak Dal Mobile app for Contactless crew sign-on/signoff, Online recharge of suburban cards, Kolkata Metro, facilitation of Payments through use of robust Digital IT enabled system with minimal staff presence etc.

Figure 2.5: Train Control and Management Systems

2.3.3 Collaborations with International Rail Safety Standards

Research, standards, and analysis are essential to ensure that innovative and sustained technology is operated safely, efficiently, and cost-effectively across the rail industry. At the rail Safety and Standards Board, they are providing independent advice, insights, and guidance to improve safety and increase efficiency in rail.

It understands the challenges of operating infrastructure and assets in a constantly changing environment. That's why it will **continue to work independently and collaboratively with operators and innovators to prepare for new developments and find the best route forward.**

Collaboration is the key and over the past year, the rail industry has come together to develop the Rail Safety Strategy aimed at delivering the shared vision of a healthier, safer, and affordable railway. This strategy covers some key risks: -

- **Public Behavior**
- **Health and wellbeing**
- **Operations**
- **Asset Management**
- **Occupational health and safety**

2.4 Limitations and Challenges of Current Systems in Indian Context

Railway accidents can **often result in serious injuries or fatalities, and railway authorities must work quickly to rescue and recover victims**. This can be a difficult and challenging task, as trains are often large and heavy, and the wreckage can be spread over a wide area.

Railway authorities must also investigate the cause of the accident in order to prevent future accidents. This can be a complex and time-consuming process, as railway accidents can be caused by a variety of factors, including human error, mechanical failure, and weather conditions.

Railway authorities must also manage public relations surrounding a train accident. This can be a delicate task, as railway authorities must balance the need to provide information to the public with the need to protect the privacy of victims and their families.

Railway authorities must also work to restore service as quickly as possible after a train accident. This can be a difficult task, as railway lines must be inspected for damage, and trains may need to be rerouted or replaced.

Railway authorities are committed to **safety, and they work hard to prevent train accidents.** However, when accidents do occur, railway authorities are prepared to deal with the challenges in order to protect the public and restore service. **[4]**

Figure 2.6: Balasore train collision, which occurred on June 2, 2023, in Balasore, Odisha, India.

Chapter 3: Fundamentals of Head on Collisions

3.1 : Understanding Head-On Collisions in Indian Railway Systems

3.1.1 : Characteristics and Frequency Analysis

Head-on collisions are among the most severe types of accidents in Indian railway systems, occurring when two trains traveling in opposite directions collide on the same track. These collisions are characterized by:

- **High Impact Forces:** The combined speed and weight of the trains result in significant impact forces.
- **Severity:** They often result in catastrophic damage, high fatality rates, and numerous injuries.
- **Frequency:** While less frequent than derailments or rear-end collisions, head-on collisions have severe consequences when they occur.
- **Types of Trains Involved**: Both passenger and freight trains can be involved, with passenger trains typically resulting in higher casualty figures.

3.1.2 : Contributing Factors and Root Causes

Several factors contribute to head-on collisions in Indian railways, including:

- **Human Error:** Mistakes made by train drivers, signal operators, or maintenance personnel.
- **Technical Failures:** Malfunctions in signaling equipment, track defects, or communication systems.
- **Organizational Issues:** Inadequate training, poor safety protocols, and insufficient oversight.
- **Communication Breakdown**: Failures in communication between train control centers and train operators.
- **Infrastructure Deficiencies:** Aging infrastructure that may not support modern safety requirements.
- **Environmental Factors:** Adverse weather conditions affecting visibility and track conditions.
- **Operational Complexity:** High traffic density and complex train scheduling increasing the risk of human and technical errors.

3.1.3 : Impact on Passenger Safety and Railway Operations

Head-on collisions have a profound impact on:

- **Passenger Safety:** High fatality rates, severe injuries, and psychological trauma for survivors.
- **Railway Operations:** Significant disruption to services, long-term service interruptions, and logistical challenges in managing accident aftermath.
- **Economic Costs:** High costs associated with repairs, compensation, and legal liabilities.
- **Public Perception:** Erosion of public trust in railway safety and reliability.
- Emergency Response: Strain on emergency services and resources during rescue and recovery operations.
- **Infrastructure Damage:** Extensive damage to tracks, signals, and rolling stock, requiring substantial repairs and downtime.

3.2 : Regulatory Framework and Safety Standards in Indian Railways

3.2.1 : Railway Board Regulations and Guidelines

The Railway Board establishes comprehensive regulations and guidelines to ensure safety, including:

- **Operational Protocols:** Detailed procedures for train operations, signaling, and communication.
- **Maintenance Standards:** Regular maintenance schedules and standards for tracks, trains, and signaling equipment.
- **Safety Training:** Mandatory training programs for all railway personnel.
- **Emergency Preparedness:** Protocols for emergency response and disaster management.
- **Inspections and Audits:** Routine inspections and audits to ensure compliance with safety standards.
- **Incident Reporting:** Mandatory reporting and investigation of all accidents and near-misses.

3.2.2 : Research and Development Initiatives for Safety Enhancement

Indian Railways engages in continuous research and development to enhance safety, focusing on:

- **Advanced Technologies:** Implementation of automatic train protection systems, collision avoidance technologies, and real-time monitoring systems.
- **Predictive Maintenance:** Use of data analytics and AI to predict and prevent equipment failures.
- **Training Simulators:** Development of advanced training simulators for train operators and signal personnel.
- **Innovative Materials:** Research into durable and resilient materials for tracks and rolling stock.
- **Human Factors:** Studies on reducing human error through ergonomic designs and improved work environments.
- **Pilot Projects:** Testing new technologies and safety measures on select routes before wider implementation.

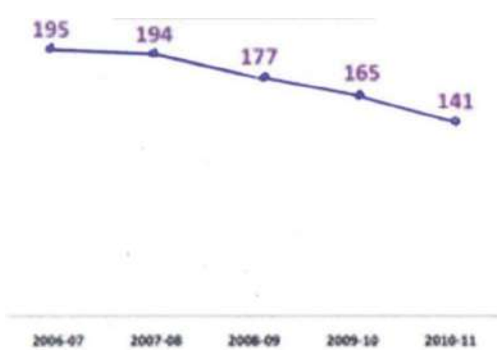
3.2.3 : Compliance with International Standards and Best Practices

Indian Railways strives to comply with international safety standards and best practices, including:

- **Global Benchmarks:** Adopting safety benchmarks from leading railway systems worldwide.
- **Collaborations:** Partnering with international railway organizations for knowledge exchange and joint research projects.
- **Certifications:** Obtaining certifications from international safety bodies.
- **Training Programs:** Sending personnel to international training programs and conferences.
- **Best Practice Implementation:** Regularly updating safety protocols to align with global best practices.
- **Safety Culture:** Promoting a culture of safety through continuous learning and improvement.

3.3: Case Studies and Analysis of Previous Incidents in Indian Railways

Analyzing previous incidents helps identify lessons and implement improvements. Key case studies include:
Firozabad Rail Disaster (1995): Analysis of signal failures, human errors, and emergency response shortcomings.
Gaisal Train Collision (1999): Examination of communication breakdowns and inadequate safety protocols.
Awadh-Assam Express and Brahmaputra Mail Collision (2010): Focus on signal misinterpretation and operational complexities.
Sainthia Train Collision (2010): Investigation into driver fatigue, signaling errors, and equipment failure.
Jnaneswari Express Derailment (2010): Study of sabotage, response times, and recovery efforts.
Kacheguda Train Collision (2019): Review of automatic train protection system failures and manual override issues.



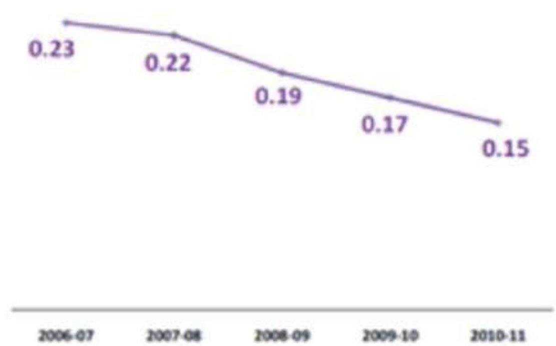


Figure3.1: Total consequential train accidents

Figure3.2: Accidents per million km

By studying these incidents, Indian Railways can:

- Identify Patterns: **Recognize common causes and factors in head-on collisions.**
- Implement Improvements: **Introduce changes to protocols**, training, and technology to prevent similar accidents.
- Enhance Training: **Develop targeted training programs** based on real-world scenarios.
- Improve Response: **Enhance emergency response strategies and coordination.**
- Monitor Compliance: **Ensure strict adherence to updated safety regulations and standards. [4]**

CHAPTER 4: FACTORS INFLUENCING HEAD-ON COLLISIONS

4.1 : Human Factors in Train Operations in the Indian Context

4.1.1 : Driver Behavior and Training Programs

The way train drivers behave and how well they are trained is very important for safety:

- **Training Programs:** Drivers need regular and good training to handle different situations. This includes practice sessions, safety drills, and tests.
- **Behavior Monitoring:** Watching how drivers work can help find risky actions and fix them. Training that focuses on good behavior can improve safety.
- **Skill Improvement:** Drivers should keep learning new skills and stay updated with the latest safety practices

4.1.2 : Crew Coordination and Communication Protocols

Good teamwork and clear communication among train crew members are essential:

- **Standard Procedures:** Clear instructions and rules help all crew members know their roles.
- **Communication Systems:** Reliable tools like radios help crew members and control centers stay in touch.
- **Team Training:** Training together helps crew members work better as a team, especially during emergencies.

4.1.3 : Fatigue Management and Work Hour Regulations

Managing tiredness among train operators is crucial:

1. **Work Hours:** Following strict rules about how long staff can work helps them stay rested.
2. **Fatigue Programs:** Programs that teach staff about the dangers of fatigue and promote good sleep habits can reduce risks.
3. **Monitoring Alertness:** Using systems to check if staff are alert and adjusting schedules based on their tiredness helps maintain safety.

4.2 : Environmental Factors Affecting Railway Safety in India

4.2.1 : Weather Conditions and Monsoon Challenges

India's weather can create challenges for railway safety:

1. **Monsoon Impact:** Heavy rains can flood tracks, cause landslides, and reduce visibility, increasing accident risks.
2. **Weather Monitoring:** Advanced systems can give real-time updates and warnings, helping to take safety measures in advance.
3. **Infrastructure Design:** Building tracks and drainage systems to handle extreme weather can reduce weather-related risks.

4.2.2 : Track Geometry and Infrastructure Maintenance

Keeping tracks and infrastructure in good condition is key:

- **Regular Checks:** Routine inspections and maintenance of tracks, bridges, and tunnels help spot and fix problems early.
- **Modern Techniques:** Using advanced tools like ultrasonic testing improves maintenance accuracy and efficiency.
- **Continuous Investment:** Regular funding for infrastructure upgrades keeps the railway system safe.

4.2.3 : Signal Systems and Visibility Issues

Good signaling systems are essential for safe train operations:

1. **Signal Maintenance:** Regular checks and timely upgrades of signals prevent failures.
2. **Visibility:** Making sure signals are visible in all weather conditions with reflective materials and proper lighting helps prevent accidents.
3. **Technology Integration:** Using advanced technologies like automatic train control systems can reduce manual errors and improve safety.

4.3 : Mechanical Factors and Equipment Reliability in Indian Locomotives

4.3.1 : Locomotive Design and Maintenance Practices

The design and upkeep of locomotives are crucial for reliability and safety:

- **Strong Design:** Locomotives should be built to handle tough conditions and heavy use.

- **Preventive Maintenance:** Regular maintenance schedules help fix issues before they become big problems.
- **Overhauls:** Major check-ups at regular intervals extend the life of locomotives and keep them reliable.

4.3.2 : Brake Systems and Emergency Controls

Reliable brakes are vital for stopping trains safely:

- **Brake Testing:** Regular testing ensures brakes work well when needed.
- **Emergency Controls:** Advanced emergency braking systems and training for crew members on how to use them can prevent accidents.
- **Backup Systems:** Having backup braking systems ensures safety even if one system fails.

4.3.3 : Track Circuits and Signaling Devices

Track circuits and signals help manage train movements safely:

- **Track Circuit Reliability:** Reliable track circuits help detect train positions accurately.
- **Advanced Signals:** Using modern signal devices with fail-safe features improves safety.
- **Continuous Monitoring:** Regular checks and real-time monitoring help find and fix faults quickly.

4.4 : Socio-Economic Impacts and Public Perception

Accidents like head-on collisions have big effects on society and how people view railway safety:

- **Economic Costs:** Accidents cause large financial losses due to repairs, compensation, and disrupted services.
- **Public Trust:** Frequent accidents can make people lose trust in the railway system.
- **Safety Investments:** Spending on safety measures and new technologies can prevent accidents and boost public confidence.
- **Community Engagement:** Keeping the public informed and involved in safety efforts can improve their trust and cooperation.

By addressing these factors, Indian Railways can make train travel safer, reduce the chances of head-on collisions, and ensure smooth and reliable operations.

CHAPTER 5: Considerable Factors for Head on collision prevention system

The Automated model of Head on Collision Prevention System for Vehicles locomotive: This automated safety device mainly relies on multiple sensor technology. The sensors continuously monitor the surrounding environment. The sensors we used here provide the real time data about the position speed and trajectory of nearby vehicles.
The data we collected from the sensors helps us to provide a collision prediction algorithm andanalyses the vehicle movement patterns and make decision. If the algorithm detects a threat, it triggers a warning braking is applied to prevent the accident.

5.1: Factors to be considered for head on collision prevention system:

5.1.1 : Real-time Data Processing:

Real time data processing is very important in head on collision prevention systems to ensure rapid detection and response to the dangers on the road.

Let's get some brief idea about the real time data operation,

a) **Data Acquisition:** Various sensors that we attached in the model collect the data in real time from the model surroundings.
b) **Detect the object:** Our algorithm analyse the data to detectthe object in front of the model and take the further actions.
c) **Collision Risk Assessment:** If the system detects any object, the system then access the risk of a collision based on the factors such asvehicles distance, vehicles speed and position.
d) **Decision making:** Based on the risk assessment the system decides what action should be taken to prevent the head on collision.

5.1.2 : Data Encryption and Authentication:

a) Data encryption and authentication are crucial components for collision prevention and actuators. This can be achieved using protocols TLS(Transport layer security)to encrypt the dataand ensures its integrity.
b) End to end encryption: The data exchanged between different components withinthe collision prevention system, including sensor data control commands must be encrypted using strong encryption algorithm. That's why end encryption is needed.
c) Authentication Mechanisms: To reduce unauthorized access to critical Components of the system, authentication mechanism is needed.

5.1.3 : Speed and distance calculation:

In head on collision prevention system, it is necessary to calculate the speed and distance between two vehicles, to avoid the head on collision and for implanting perfect implementing measures.

a) **Sensor and data fusion:** The system uses the data from various sensor that are attached in the vehicle model and gather information about the vehicle surroundings. The sensors which we attached in vehicle used to provide various types data altogether.
b) **Speed measurement:** The sensor (lidar sensor) we used in front of our vehicle model is used to measure the speed of the upcoming vehicle. In lidar sensor we use radio waves for to detect some object near to calculate the relative speed between the host vehicle and other objects we need to compare the frequency or time delay of the reflected signals.
c) **Distance Estimation:** The distance measurement sensor estimate the distance between the host vehicle and upcoming vehicle. Radar and lidar sensors measure the time it takes for emitted signals to bounce off objects and return to the sensor, allowing for precise distance estimation.
d) **Collision Risk Assessment:** By using the calculated speed, distance, information the system assesses the risk of head on collision.
e) **Warning and Intervention**: If the system detects a high risk of a head-on collision, it can alert the driver through visual or auditory warnings to take evasive action. In more advanced systems, automatic interventions such as automaticemergency braking or steering assist may be activated to mitigate the collision risk.

5.1.4 Localization and Mapping:

Localization and mapping playsa crucial role in head on collision prevention system.

a) Global Positioning System:

GPS is generally used for initial localization,it provides the vehicle's global coordinates. While GPS gives global positioning information, it may not be precise enough for urban environments or areas with high rise buildings where satellite signals may be reached. Beside this it can provide a rough estimate of the vehicle's location.

b) Inertial Measurement Units (IMUs):

IMUs consisting of accelerometers and gyroscopes provide continuous measurements of the vehicle's acceleration and orientation. By integrating these measurements over time, IMUs can estimate the vehicle's trajectory and position. However, IMUs suffer from drift over time, so they are often used in conjunction with other localization methods to correct for errors.

c) Lidar-based Localization:

Lidar sensors emit laser beams and measure the time it takes for the beams to reflect off objects in the environment.

By comparing these measurements with a pre-existing map of the environment, lidar-based localization algorithms can determine the vehicle's position and orientation. Lidar provides accurate range measurements and works well in various lighting conditions.

5.1.5 Obstacle Detection and Classification:

It is very important to detect and classify the obstacle in head on collision prevention system.

a) **Sensor fusion**:
We attached many sensors such as radar, lidar sensorand ultrasonic sensors to gather information about the surroundings of the vehicle. By sensor fusion technique we integrate the information and understand the environment.

b) **Obstacle detection:**
The sensors attached with the vehicle continuously scan the environment and detect any obstacle in its path.

c) **Distance and Speed estimation:**
The system estimates the distance and relative speed of the detected obstacles to assess the risk of collision. This information is crucial for determining the urgency of evasive manoeuvres.

5.1.6 Emergency Brake Control:

Emergency Brake control is very important for head on collision prevention system.

1. **Collision Risk assessment:**
Before applying the brake, the system assessthe severity of the potential collision based on factors such as the relative speed and distance between the vehicles, their trajectories, and the available braking distance.
2. **Precharge:**
To anticipate a potential collision the system may precharge the vehicle's brake system, applying partial braking force to reduce the response time when emergency braking is initiated.
3. **Collision Warning and Automatic Emergency Braking(AEB):**
The system issues a auditory warning alert to the driver before engaging emergency brake. If the system determines that a collision is imminent and the driver has not taken sufficient action to avoid it, automatic emergency braking is initiated.

5.2 The prevention system to be performed:

5.2.1 Locomotive sensing and Automatic breaking system

Ensuring passenger and cargo safety in railway transportation is crucial. Modern locomotives use advanced sensing and braking systems **to detect obstacles, monitor track conditions, and apply brakes promptly, ensuring smooth and safe operations.**

Our model incorporates key components for enhanced safety and reliability: a 3.7V 2000mAh Li-ion rechargeable battery, L298N Motor Driver Shield, high-torque Johnson DC gear motor with wheel, and Sharp IR sensor (GP2Y0A21YK0F). **The Sharp IR sensor is chosen** for its long detection range (20-80 cm), enabling accurate obstacle and track condition detection.

The high-torque DC gear motor powers and brakes the locomotive, while the L298N Motor Driver Shield manages motor control and power. The system is powered by high- energy-density 3.7V Li-ion batteries, minimizing downtime for recharging.

The setup involves connecting four high-torque DC gear motors to two L298N Motor Drivers. The motor drivers communicate with an Arduino for direction and speed control. The Sharp IR sensor connects to the Arduino for obstacle detection. The motors are powered by three 3.7V Li-ion batteries, and a 9V battery powers the Arduino.

5.2.2 Accident prevention signal

Our model uses an accident prevention system with **two ultrasonic sensors and an Arduino to detect locomotives on the same track and measure their speed. Here's a concise summary:**

Detection: Two ultrasonic sensors are placed on the track. They send out pulses and measure the time it takes for the echo to return, calculating the distance to any object (locomotive).

Speed Measurement: When a locomotive passes both sensors, the time difference between detections is used to calculate its speed.

Signal: If the calculated speed exceeds 0.5 km/h, an LED indicator turns on, signaling that another locomotive is on the track. The system resets after 30 seconds.

Monitoring: If no locomotive is detected, the system prints "No Locomotive detected" and rechecks after 1 second. If only one sensor is triggered, it prints "Searching..." and waits for the second detection.

5.2.3 : Accident alert system

The primary objective of this model is to use **an MPU6050 accelerometer and gyroscope sensor to detect abnormal motion**, such as an accident, and trigger an alert using a buzzer or LED, **while also notifying the nearest control room.**

How It Works:

Continuous Monitoring: The MPU6050 sensor continuously monitors motion.
Data Processing: Sensor data is read and processed every second.
Threshold Detection: The system checks if motion values exceed predefined thresholds, indicating an accident.
Alert Activation: If abnormal motion is detected, the alert system (buzzer or LED) is activated, and an emergency message is sent to the control room.
System Components:

Sensor Setup: Connect the MPU6050 to an Arduino.
Data Normalization: Convert raw sensor data to a usable format.
Threshold Setting: Define abnormal motion thresholds for each axis (X, Y, Z).
Alert Mechanism: Trigger alerts if motion exceeds thresholds.
Error Handling: Implement error checks for sensor reliability.
Testing and Calibration: Ensure sensor accuracy through testing and calibration
.
This model ensures continuous monitoring for sudden changes in motion, making it suitable for applications requiring real-time accident detection and alerting.

CHAPTER 6: THE HEAD ON COLLISION PREVENTION MODEL (LOCOMOTIVES):

6.1 THE COLLISION PREVENTION MODEL

Our automated head on collision prevention model will be an embedded system for automatic control for preventing head on collision between locomotives and implementing new age technologies using various sensors, gps, cellular module to monitor the real time location of trains and to prevent any types of accidents for providing safety of thousands of people.
We have already discussed the automations that need to be done in the previous section. As a result, we will begin our suggested model by addressing each of those factors individually in the manner shown below.

6.1.1 Basic Components to be used

It is evident that there are certain common components that will be utilized in every system.
These are as follows:

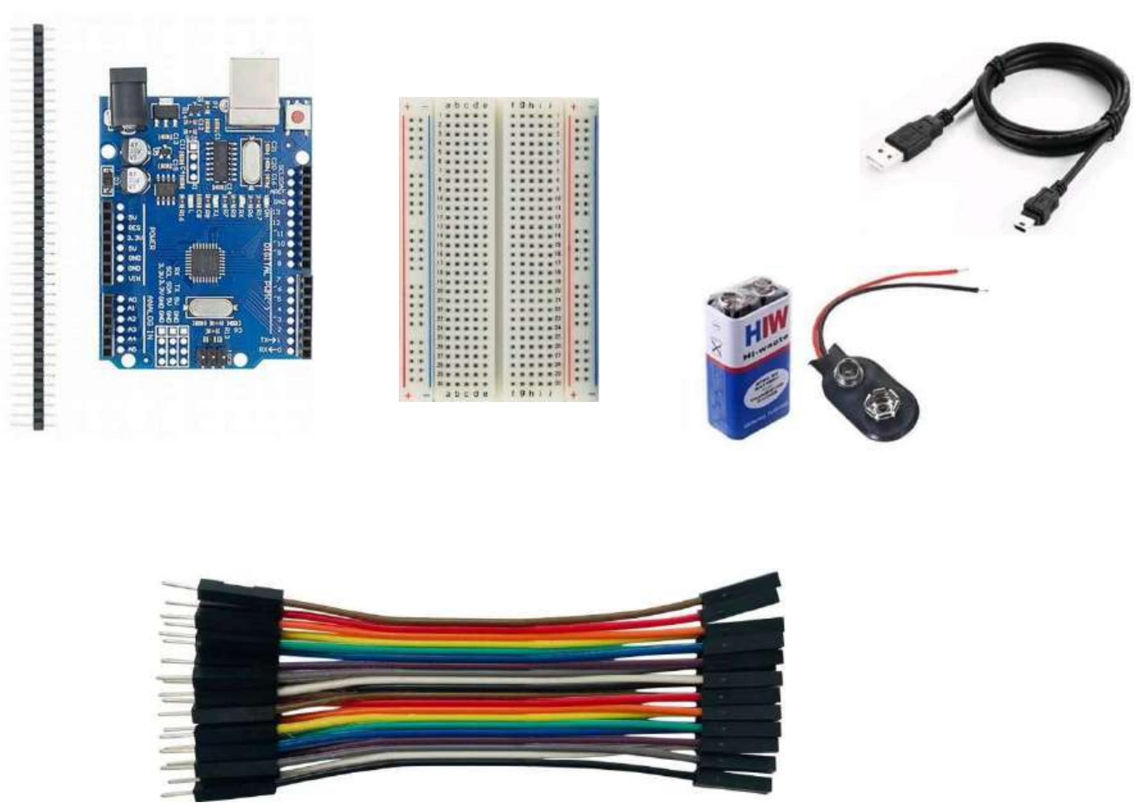


Figure 6.1: Some of the Basic Components to be Used

6.1.1.1 Arduino UNO

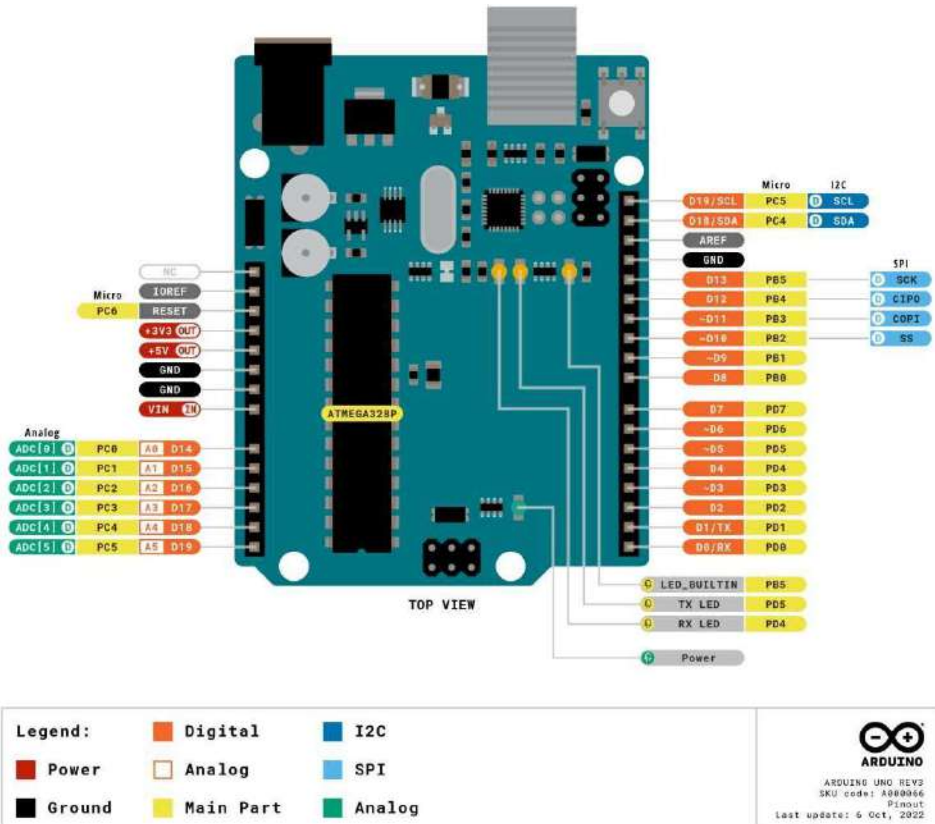


Figure 6.2: Arduino UNO and its Pin Configuration

Arduino UNO is a microcontroller board based on the ***ATmega328P***. It has ***14 digital input/output pins*** (of which 6 can be used as PWM outputs), ***6 analog inputs***, a *16 MHz* ***ceramic resonator***, a ***USB connection***, a ***power jack***, an ***ICSP header*** and ***a reset button***. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with an AC-to-DC adapter or battery to get started.

The ATmega328P also features ***1kb of EEPROM***, a memory which is not erased when powered off. The ATmega328P can easily be replaced, as it is not soldered to the board. The Arduino UNO features a ***barrel plug connector*** that works great with a standard 9V battery. **[6]**

Table 6.1: the Technical specifications for the Arduino UNO R3

Board	Name	Arduino UNO R3
	SKU	A000066
Microcontroller	ATmega328P	
USB connector	USB-B	
Pins	Built-in LED Pin	13
	Digital I/O Pins	14
	Analog input Pins	6
	PWM pins	6
Communication	UART	Yes
	I2C	Yes
	SPI	Yes
Power	I/O Voltage	5V
	Input voltage(nominal)	7-12V
	DC Current per I/O Pin	20mA
	Power Supply Connector	Barrel Plug
Clock Speed	Main Processor	ATmega328P 16 MHz
	USB-Serial Processor	ATmega12U2 16 MHz
Memory	ATmega328P	2KB SRAM, 32KB FLASH, 1KB EEPROM
Dimensions	Weight	25g
	Width	53.4mm
	Length	68.6mm

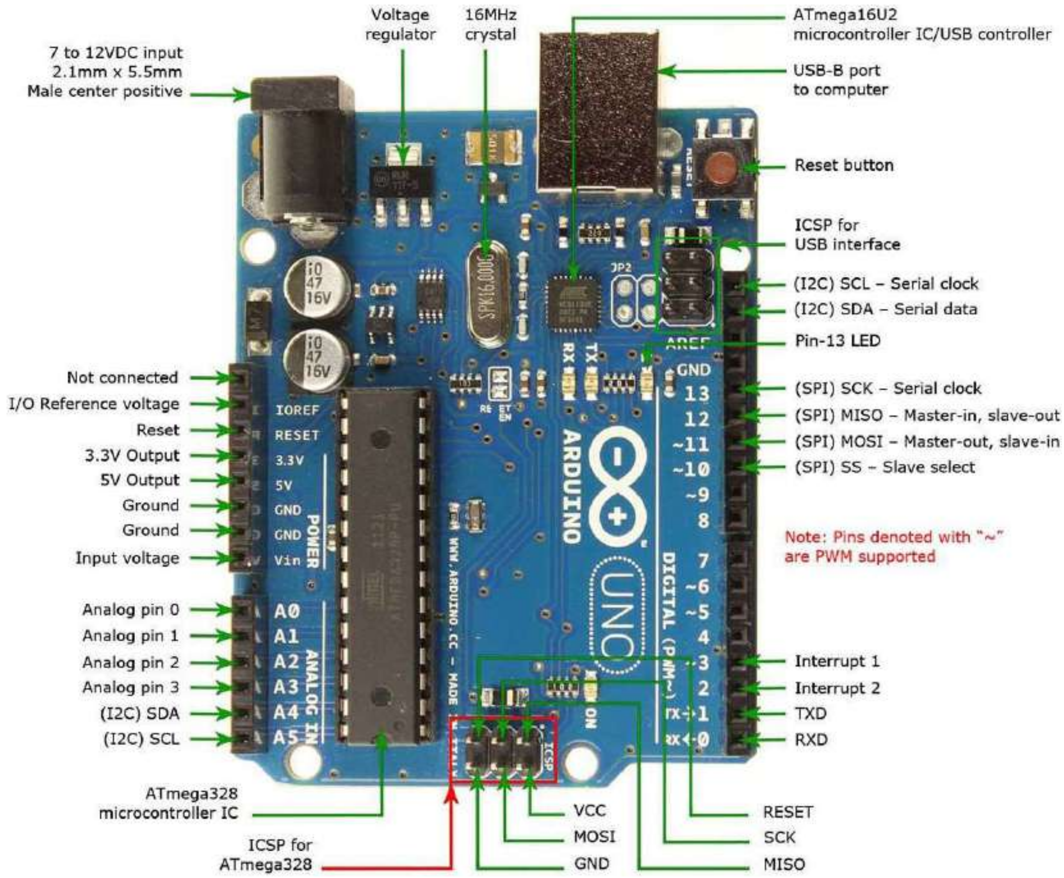


Figure 6.3: Different Parts and Components of Arduino UNO

6.1.1.2 Jumper Wires

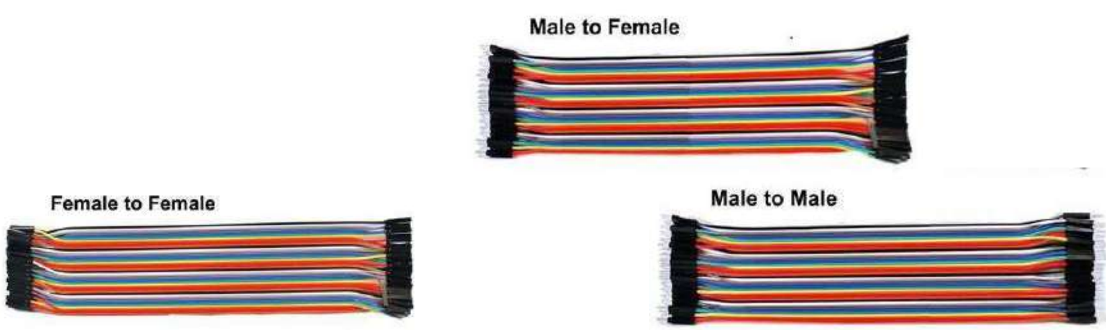


Figure 6.4: Female to Male, Male to Male and Female to Female Jumper Wires

Generally, jumpers are tiny metal connectors used to **close or open a circuit part**. They have **two or more connection points**, which regulate **an electrical circuit board**. Their function is to **configure the settings for computer peripherals**, like the **motherboard**. Suppose our motherboard supported intrusion detection. A jumper can be set to **enable or disable** it.

Jumper wires are electrical wires with connector pins at each end. They are used to connect two points in a circuit without soldering.We can use jumper wires to **modify a circuit or diagnose problems in a circuit.** Further, they are best **used to bypass** a part of the circuit that does not contain a resistor and is suspected to be bad. This includes a stretch of wire or a switch. Suppose all the fuses are good and the component is not receiving power; find the circuit switch. Then, bypass the switch with the jumper wire.

How much current (I) and voltage (V) can jumper wires handle? The I and V rating will depend on the copper or aluminum content present in the wire. For an Arduino application is no more than **2A** and **250V**. We also recommend using solid-core wire, ideally **22 American Wire Gauge** (***AWG***). **[7]**

Types of Jumper Wires

Common Connector Standard: The **male-to- male** configuration ensures compatibility across various electronic components, providing a standardized connection approach.

Figure 6.5: the Male- Male jumper wires

Male-to-Female Dynamics: This setup enables the connection of male pins to female receptacles, offering flexibility in interfacing with a range of components.

Figure 6.6: the Male- Female jumper wires

Female-to-Female Flexibility: These wires provide a unique solution for connecting components with female connectors, broadening the compatibility spectrum in circuit design.

Figure 6.7: the Male- Female jumper wires

Here in our project, we have used jumper wires (M-M, M-F, F-F) and pin headers as connectors for different purpose

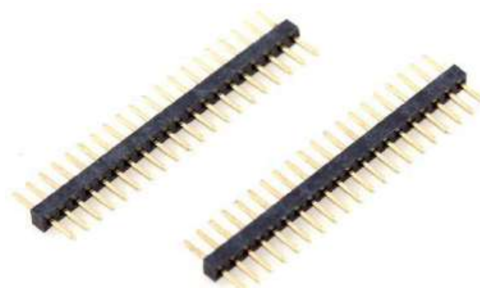
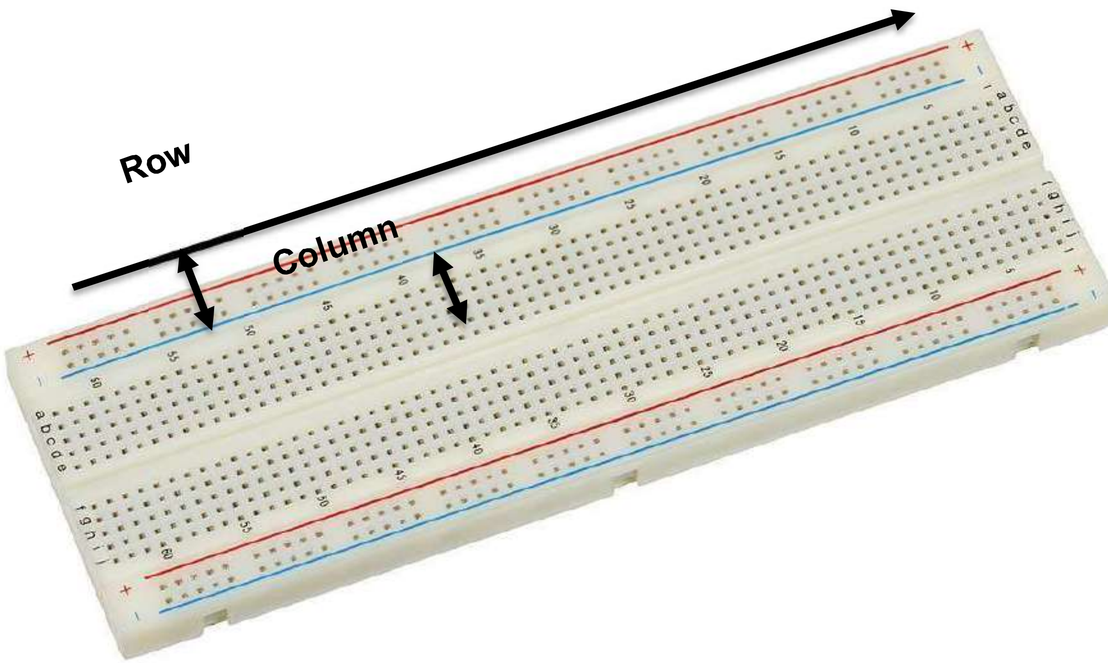


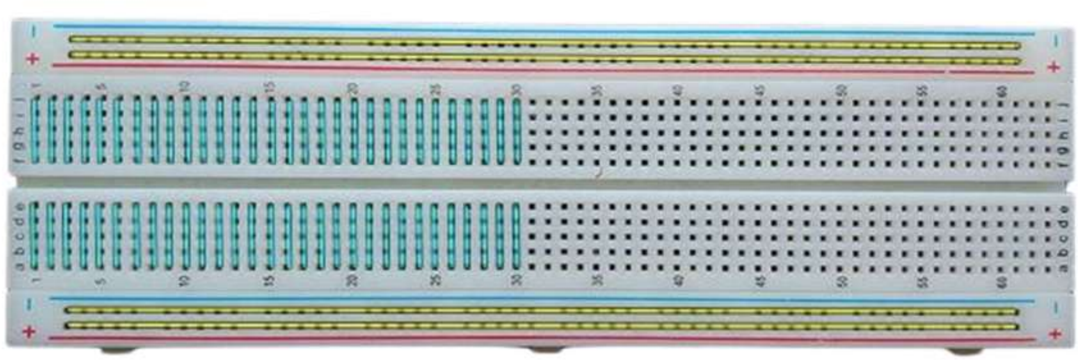
Figure 6.8: Pin Headers as connectors

A **pin header** (or simply header) is a form of electrical connector. A male pin header consists of one or more rows of metal pins molded into a plastic base, often **2.54 mm** (**0.1 in**) apart, though available in many spacing. Male pin headers are cost-effective due to their simplicity. The female counterparts are sometimes known as **female headers** (also as **pin sockets**), though there are numerous naming variations of male and female connectors. Historically, headers have sometimes been called "**Berg connectors** "or "**DuPont**" connectors, but headers are manufactured by many companies.

6.1.1.3 Breadboard

A Breadboard is a tool for easily connecting wires and other electronic elements when prototyping circuits. The Breadboard has two main sections - the outside rows and the middle columns. The holes or contact points in the row sections are only electrically connected to contact points in the same row. The contact points in the column sections are only electrically connected to the contact points in the same column (see Figure). **Every hole within each row is interconnected, and every hole within each column is interconnected.**





Connected Vertically

Connected Horizontally

Figure 6.9: Internal Circuit Diagram of Breadboard

A modern solderless breadboard socket consists of a **perforated block of plastic with numerous tin plated phosphor bronze or nickel silver alloy spring clips** under the perforations. The clips are often called **tie points or contact points**. The number of tie points is often given in the specification of the breadboard. The spacing between the clips (lead pitch) is typically **0.1 inches (2.54 mm)**. **Integratedcircuits (ICs) in dual in-line packages (DIPs)** can be inserted to straddle the centerline of the block. Interconnecting wires and the leads of discrete components (such as capacitors, resistors, and inductors) can be inserted into the remaining free holes to complete the circuit.

Where ICs are not used, **discrete components and connecting wires may use any of the holes**. Typically the spring clips are rated for **1 ampere at 5 volts and 0.333 amperes at 15 volts (5 watts)**.Solderless breadboards connect pin to pin by **metal strips inside the breadboard**. The layout of a typical solderless breadboard is made up from two types of areas, called strips. Strips consist of interconnected electrical terminals. Often breadboard strips or blocks of one brand have male and female dovetail notches so boards can be clipped together to form a large breadboard. **[9]**

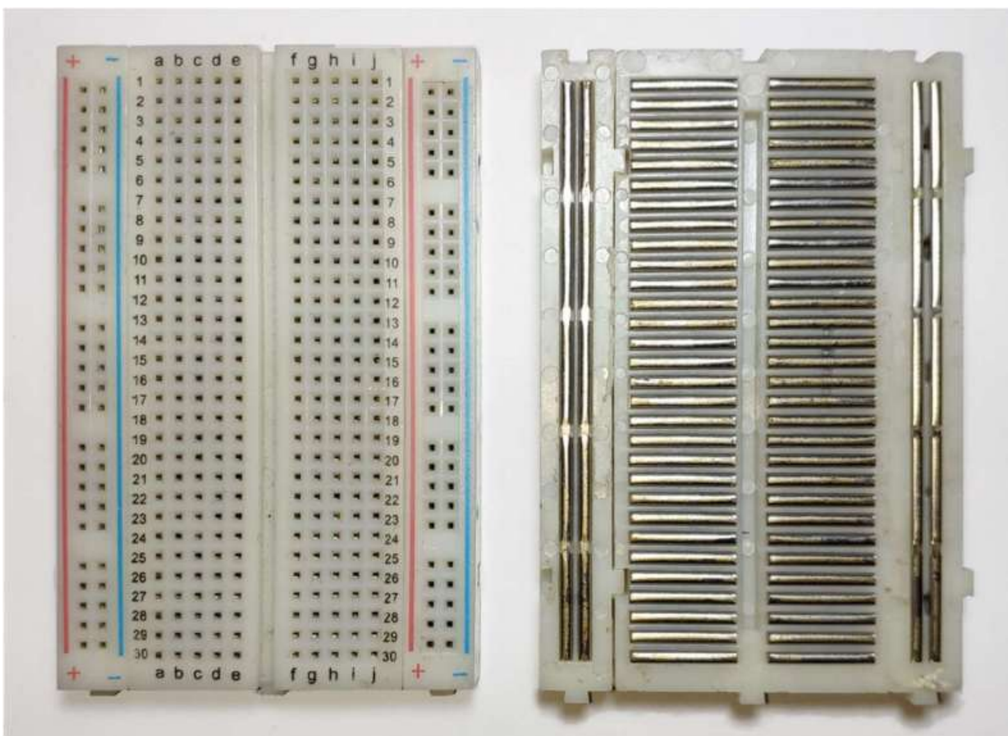


Figure 6.10: External Look and Internal Circuitry of Breadboard

The main areas, to hold most of the electronic components, are called **terminal strips**. In the middle of a terminal strip of a breadboard, one typically finds a notch running in parallel to the long side. The notch is to mark the centerline of the **terminal strip and provides limited airflow (cooling) to DIP ICs straddling the centerline**. The clips on the right and left of the notch are each connected in a radial way; typically, five clips (i.e., beneath five holes) in a row on each side of the notch are electrically connected. **The five columns on the left of the notch are often marked as A, B, C, D, and E, while the ones on the right are marked F, G, H,I and J**. When a "skinny" dual in-line pin package (DIP) **integrated circuit (such as a typical DIP-14 or DIP-16, which have a 0.3-inch (7.6 mm) separation between the pin rows) is plugged into a breadboard,** the pins of one side of the chip are supposed to go into column E while the pins of the other side go into column F on the other side of the notch.

The rows are identified by numbers from 1 to as many as the breadboard design goes. A full-size terminal breadboard strip typically consists of around **56 to 65 rows of connectors.** Together with bus strips on each side this makes up **a typical 784 to 910 tie point solderless breadboard**. Most breadboards are designed to accommodate **17, 30 or 64 rows in the mini, half, and full configurations respectively**. **[9]**

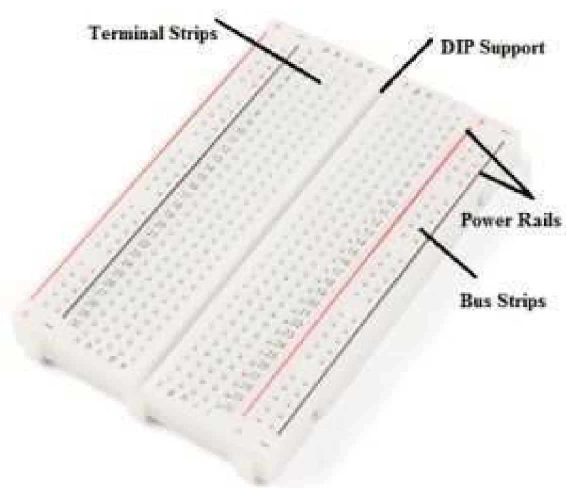


Figure 6.11: Breadboard Layout

To provide power to the electronic components**, bus strips are used.** A bus strip usually contains **two columns: one for ground and one for a supply voltage.** However, some breadboards only provide a single-column power distribution bus strip on each long side. Typically, the row intended for a **supply voltage is marked in red, while the row for ground is marked in blue or black.** Some manufacturers connect all terminals in a column. Others just connect groups of, for example, 25 consecutive terminals in a column. The latter design provides a circuit designer with some more control over crosstalk (inductively coupled noise) on the power supply bus.

Often the groups in a bus strip are indicated by gaps in the color marking. Bus strips typically run down one or both sides of a terminal strip or between terminal strips. On large breadboards additional bus strips can often be found on the top and bottom of terminal strips.
Some manufacturers provide separate bus and terminal strips. Others just provide breadboard blocks which contain both in one block.

Advantages of Breadboard:

Temporary Connections: One of the most compelling reasons to use a breadboard is its ability to create temporary electrical connections. Unlike soldering, which is permanent and can be challenging to modify, breadboards allow you to quickly and easily make and remake connections as needed.

No Soldering Required: Soldering can be intimidating for beginners, as it requires specific tools and skills. Breadboard eliminate the need for soldering entirely, making electronics experimentation accessible to a broader audience.

Component Reusability: On a breadboard, you can reuse components without causing damage. This feature is particularly valuable when working with expensive, hard-to-find, or rare components, as you can use them in multiple projects.

Error-Friendly: Mistakes are an inherent part of the learning and design processin electronics. Breadboard accommodates errors gracefully. If a circuit doesn'tperform as expected, you can easily identify and rectify the problem without damaging any components.

Rapid Prototyping: Breadboards enable rapid prototyping, allowing you to iterate and refine your circuit designs quickly. This speed is particularly crucial in industries where time-to-market is a critical factor.

6.1.1.4 USB Cable for Arduino UNO (Printer Cable)

Figure 6.12: A to B Type Male to Male USB

A printer cable is a type of computer cable used for connecting a printer to a computer. It is used for transmitting print commands and functions from the computer to the printer. The cable has a connector at each end. A printer cable works by connecting one end of the cable to the printer and the other end to a computer port. When a user sends a print command, it is transmitted via the printer cable from the computer to the printer, which then prints the data. The cable is also used to send back information and other messages from the printer to the computer, such as the ink level, printer and paper status, etc.
Cable for Arduino UNO is the most common A to B Male to Male type peripheral USB cable for Arduino. **[10]**

Figure 6.13: Different Types of Printer Cable

The cable has the following specifications and features:

Table 6.2: Specifications and Features of Printer Cable

Length is 50cm or 1m or 1.5m or any other size.
Hot Pluggable.
Fully compatible with the PC.
Molded strain relief and PVC over molding to ensure a lifetime of error-free data transmissions.

Foil and braid shield comply with fully rated cable specifications reducing EMI/FRI interference.
For Error-Free High-Performance Transmission.

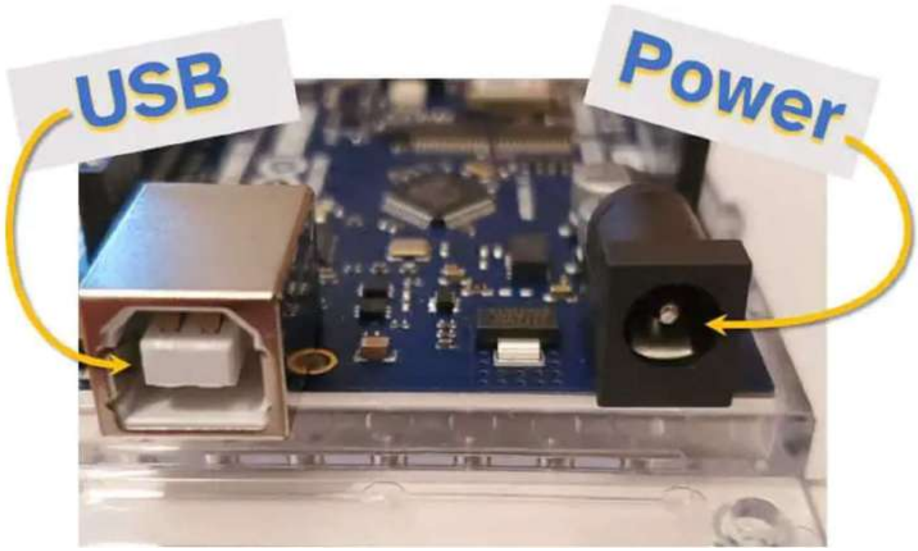


Figure 6.14: Arduino Uno with plugs for USB and Power

Figure 6.15: Arduino Uno powered from a USB phone charger using a USB Type B cable

6.1.1.5 9V battery and battery connector

Figure 6.16: 9V battery and connecter

9V Battery is the most commonly used and it is a portable 9V battery. It is non-rechargeable and is a high-capacity and low-cost solution for many electronic devices. This battery is a high-capacity & low-cost solution for many electronic devices. It is used with its specific battery snap, the battery clip can be used to power LEDs or other devices with a 9V battery. It is used for Electronic Toys, Clocks, Walkie-Talkies, Smoke Alarms, Handheld Test Equipment, Digital Instruments, Transistor Radios & General purposes.

This 9V Battery is a reliable and comprehensive power solution for electronic projects. It features advanced charge protection technology that ensures superior safety and efficiency. This battery packs enough energy to support a variety of applications, including robotics, motor controllers, sensors, and more. [11]

Table 6.3: Specifications of 9V battery

Discharge Time	**270hm, 9hours**
Nominal Voltage (V)	**9**
Jacket Material	**Metal**
Discharge Terminal Type	**Press Stud Terminal**

Dimensions (mm) LxWxH	**4.5 x 2.2 x 1.5**
Weight (gm)	**178**

A 9V Battery connector is used to hold 9V batteries on one side and on the other side, connector wires are plugged into the power supply unit to supply power to the whole circuit. Generally, we operate our robot in dc power supply. So connectedthe dc power source via a dc connector. One part of the connector is a connector of the development board and the other is connected to the dc power supply.

DC power supply has two different connectors one is male and another is female, as the same type the 9V dc battery connector have also male and female connector which is connected to the battery and complete the circuit.

Figure 6.17: 9V Battery Snap Connector with jack

Application: Use to connect the dc power supply to the development board. Provide a path between dc power supply and passive parts.

Table 6.4: Specifications of 9V Battery Snap Connector

Connector for	**9V Battery**
Colour	**Blue or Black**
Length (cm)	**14**
Weight (gm)	**1.2 approx. (each)**

There are various advantages of this 9V battery as follows:

- **Compact Size:** The compact size of 9-volt batteries makes them particularly advantageous for devices where space is limited. Their rectangular shape and small footprint allow them to fit into tight spaces within electronic devices. This feature is especially useful in applications where miniaturization is essential, such as in small electronic gadgets and medical devices.

- **Long Shelf Life:** One of the key advantages of 9-volt batteries is their relatively long shelf life when stored properly. This means that they can be purchased and kept in storage for extended periods without significantly losing their charge. As a result, they are often chosen for emergency devices like smoke detectors and emergency flashlights, where reliability over timeis crucial.

- **Wide Range of Applications:** Due to their versatility and compatibility with various electronic devices, 9-volt batteries find applications in numerous industries and products. They are commonly used in smoke detectors, carbon monoxide alarms, garage door openers, wireless microphones, guitar pedals, electronic door locks, and other portable electronic devices.

- **Easy to Install and Replace:** Another advantage of 9-volt batteries is their ease of installation and replacement. Most devices that use 9-volt batteries feature simple battery compartments or holders that allow users to quickly swap out depleted batteries for fresh ones. This ease of use is particularly beneficial in situations where rapid battery replacement is necessary, such as in emergency equipment.

- **Steady Voltage Output:** While the voltage of a battery gradually decreases as it discharges, 9-volt batteries are known for providing a relatively steady voltage output throughout their lifespan. This consistent power delivery ensures that devices powered by 9-volt batteries maintain stable performance until the battery is fully depleted.

- **Reliable Power Source:** 9-volt batteries are renowned for their reliability as a power source.

They are engineered to deliver consistent performance under various conditions, including temperature extremes and high-drain applications. As a result, they are trusted by manufacturers and consumers alike for powering critical electronic devices where uninterrupted operation is essential.

- **Low Self-Discharge Rate**: 9-volt batteries often have a low self-discharge rate, meaning they retain their charge for extended periods when not in use. This attribute makes them suitable for devices that may sit idle for long periods between uses, such as emergency equipment or backup devices.

We now begin by discussing each of the project's component systems, including their methodology, needs, and our attempts to implement them, and, finally, their prospects for the future. [14]

6.1.1.6 LED

An LED is a two-lead semiconductor light source, which emits lights when activated. When an appropriate voltage is applied to the LED terminal, then the electrons are able to recombine with the electron holes within the device and release energy in the form of photons. This effect is known as electroluminescence. **The color of the LED** is determined by the energy band gap of the semiconductor.

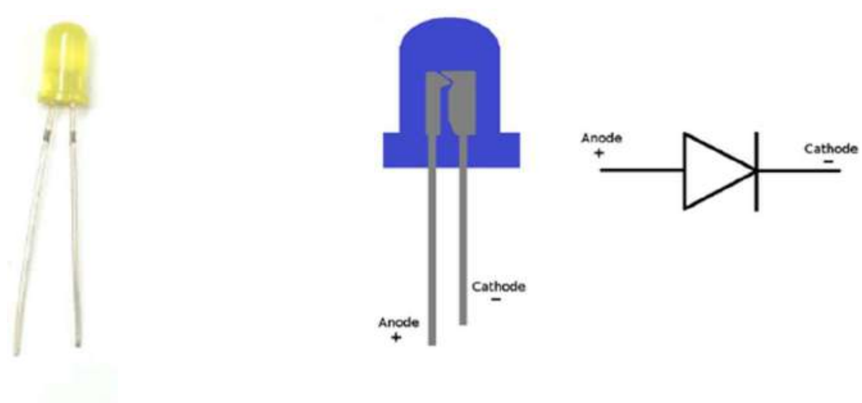


Figure 6.18: Led and its pinout

Pin Name	**Description**
Anode	Positive terminal of LED
Cathode	Negative terminal of LED

Features and Technical Specifications

- Superior weather resistance
- 5mm Round Standard Directivity
- UV Resistant Eproxy
- Forward Current (IF): 30mA
- Forward Voltage (VF): 1.8V to 2.4V
- Reverse Voltage: 5V
- Operating Temperature: -30℃ to +85℃
- Storage Temperature: -40℃ to +100℃
- Luminous Intensity: 20mc

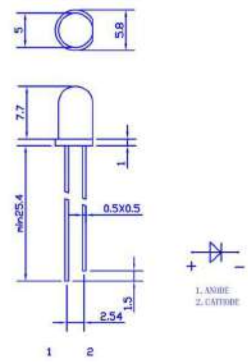


Figure 6.19: 2D model of led

6.1.1.7 Buzzer

A buzzer is a small yet efficient component to add sound features to our project/system**. It is very small and compact 2-pin structure** hence can be easily used on breadboard, Perf Board and even on PCBs which makes this a widely used component in most electronic applications.

There are two types of buzzers that are commonly available. The one shown here is a simple buzzer which when powered will make a Continuous **Beeeeeeppp.... sound**, the other type is called a readymade buzzer which will look bulkier than this and will produce a Beep. Beep. Beep. Sound due to the internal oscillating circuit present inside it. But, the one shown here is most widely used because it can be customised with help of other circuits to fit easily in our application.

This buzzer can be used by simply powering it using **a DC power supply ranging from 4V to 9V. A simple 9V battery can also be used,** but it is recommended to use a regulated +5V or +6V DC supply. The buzzer is normally associated with a switching circuit to turn ON or turn OFF the buzzer at required time and require interval.[15]

Figure 6.20: Buzzer

Buzzer Features and Specifications

- Rated Voltage: 6V DC
- Operating Voltage: 4-8V DC
- Rated current: <30mA
- Sound Type: Continuous Beep
- Resonant Frequency: ~2300 Hz
- Small and neat sealed package
- Breadboard and Perf board friendly

6.1.1.8 Slide Switch

There are different methods available to control **the flow of current in a circuit**. Among them, slide switches are most frequently used where particularly space constraints need to be taken into consideration. **These switches provide a suitable & consistent technique** through which control can be performed with circuit connection or disconnection which is achieved by a simple sliding motion of a switch.
It is a mechanical switch that is used to control the flow of current in a circuit by sliding the slider from the **OFF (open) position to the ON (close) position** known as a slide switch. This switch simply controls the current within a circuit without cutting a wire manually. **These switches will stay in one position until changed into another position manually.** The slide switch symbol is shown below.[16]

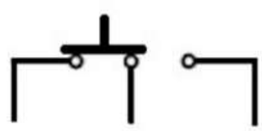


Figure 6.21: Switch Symbol

The construction of the slide switch can be done by using **metal slides that contact the plane metal elements on the switch.** When the slider in the switch is moved then metal slide contacts will slide from one set of metal contacts to the other for activating the switch.

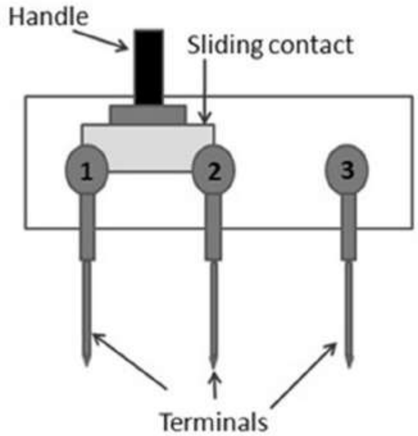


Figure 6.22: Slide Switch Construction

Slide Switch Specifications

The **specifications of the slide switc**h include the following which vary based on manufacturers.

- The max voltage across the switch or voltage rating is up to 24 Vdc.
- The max current throughout the device or switches current rating is up to 500 mA.
- The height of the actuator is flat or raised.
- The pitch or distance between pins is 2.54 mm or 5.08 mm.
- These switches can oppose dust & moisture.
- Contact rating is 12 V DC, 200 mA.
- Contact resistance is below 50 mOhms.
- Insulation resistance is above 100 MOhms @ 500 V DC.
- The strength of the dielectric is 500 V, 50 Hz for a 1-minute duration of time.
- Operating temperature ranges from -10o C to + 60o C.
- Minimum mechanical life 5000 operations.
- Contacts are silver-plated and phosphor bronze.
- Terminals are brass silver-plated.
- Contact timing is non-shorting.

6.1.2 Locomotive sensing and Automatic breaking system

6.1.2.1 APPROACH

Ensuring the safety of passengers, goods, and railway infrastructure is of utmost importance in the dynamic realm of railway transportation. In response to this requirement, contemporary locomotives are outfitted with advanced sensing and braking systems that are intended to identify obstructions, keep an eye on track conditions, and apply the brakes in a timely manner to avoid mishaps and guarantee seamless operations.

In our model, we have used carefully designed parts, as the **Li-ion rechargeable battery (3.7v 2000mah 18650)**, **Motor Driver Shield (L298N)**, **High torque DC gear motor with wheel (*Johnson DC Gear motor*)**, and **Sharp IR sensor (*GP2Y0A21YK0F*)**. Together, these elements provide unmatched safety and dependability in railway operations, serving as the foundation of a strong locomotive sensing and automated braking system.

The primary reason the GP2Y0A21YK0F model of Sharp infrared sensor is chosen for autonomous braking and locomotive sensing systems is its impressive detection range. More than its competitors, the GP2Y0A21YK0F sensor has a longer range of 20 to 80 cm, which allows for accurate identification of track conditions and impediments at a considerable distance from the locomotive.

The motor and wheel combination is made up of locomotive wheels integrated with a high-torque DC gear motor (the Johnson DC Gear motor) that may be used for both propulsion and braking. This unit provides the locomotive with the power it needs to go forward by producing the torque required and allowing for fine control over the train's direction and speed. The sturdy construction of the Johnson DC Gear motor guarantees dependable operation in railway settings, and its powerful torque output enables effective movement, even when moving steep terrain or big loads.

The locomotive's DC motor speed and direction are controlled by the Motor Driver Shield, which is particular to the L298N type. Precise motor control and effective power management are guaranteed by its integrated circuits, which enable smooth communication between the motor and the locomotive's control system. Li-ion rechargeable batteries—specifically, 3.7V 2000mAh 18650 cells—power the locomotive. The locomotive's motor and other components may be reliably powered by these very energy-dense, rechargeable batteries, which also minimize the need for downtime for recharging.

Now all the 4 high-torque DC gear motors are connected with 2 L289N Motor drivers through 5mm wires. A basic pin allocation is used by the Motor Driver Shield (L298N) to interact with an Arduino: IN1 and IN2 pins for motor direction control, and ENA for PWM-based motor speed control. As another motor is required, additional pins IN3, IN4, and ENB are used to control it. Then we connect the Vcc pin of the Sharp IR sensor (GP2Y0A21YK0F) to 5V, the GND pin to ground, and the Vo pin to an analog input pin (A0). To give power to all the 4 motors three 3.7 volt Li-Ion rechargeable batteries are connected across motordrivers' (L289N) 12V and GND pin. Another 9v disposable battery is connected to arduino to provide power to all system. Now the locomotive will run on the trackand if the sensor sensed any obstacle and another locomotive, it will stop the locomotive by sending signal to motor driver through Arduino . L289N will use its pwm pins to slowly stop the locomotive.
So, basically the functioning of the entire system now goes like the following:

- First dc motors are connected with L289N motor drivers and the wheels of the locomotives are moving.
- The arduino and Li-ion batteries are connected with motor driver for controlling and providing power.
- The sharp-IR is connected in breadboard across arduino for sensing.
- When the sharp-IR senses something like obstacle or running locomotives it will notify motor driver through arduino .
- The motor driver uses its pwm pin for barking and stop the locomotives slowly.

This entire procedure will continue the same way every time it is needed to run the locomotives and sensing and automatic braking.

6.1.2.2 COMPONENTS USED

Apart from the basic components that we have already mentioned above, we will also be using certain other components here and they are **Li-ion rechargeable battery (3.7v 2000mah 18650)**, **Motor Driver Shield (L298N)**, **High torque DC gear motor with wheel (*Johnson DC Gear motor*)**, and **Sharp IR sensor (*GP2Y0A21YK0F*).**

6.1.2.2.1 Sharp IR sensor

An IR distance sensor uses a beam of infrared light to reflect off an object to measure its distance. The distance is calculated using triangulation of the beam of light.

The sensor consists of an IR LED and a light detector or PSD (Position Sensing Device). When the beam of light gets reflected by an object, the reflected beam will reach the light detector and an ‘optical spot’ will form on the PSD.

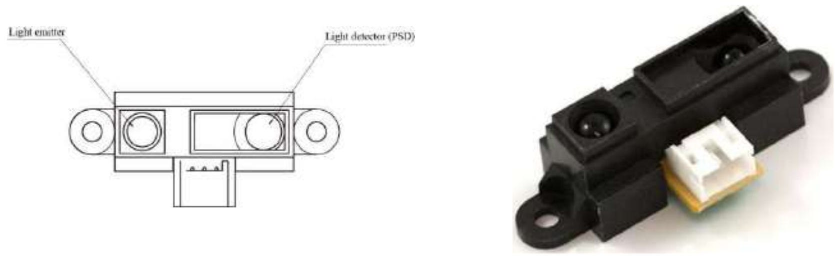


Figure 6.23: The Sharp IR sensor

When the position of the object changes, the angle of the reflected beam and the position of the spot on the PSD changes as well. See point A and point B in the image below.

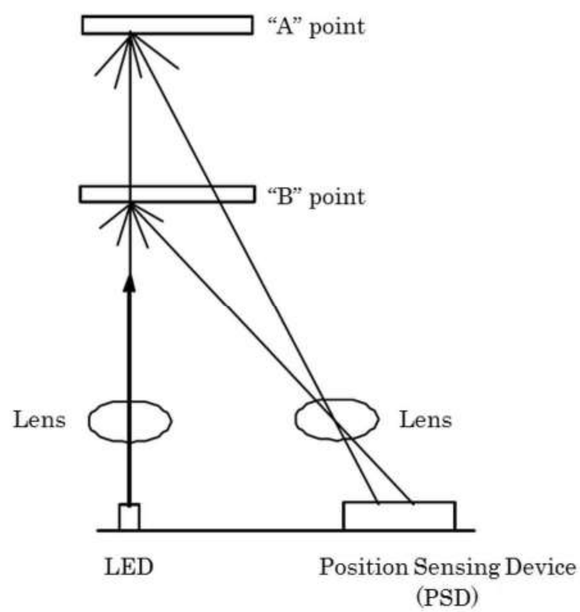


Figure 6.24: The change in the angle of the reflected beam and the position of the ‘optical spot’ on the PSD.

The sensor has a built-in signal processing circuit. This circuit processes the position of the optical spot on the PSD to determine the position (distance) of the reflective object. It outputs an analog signal which depends on the position of the object in front of the sensor.[17]

IR distance sensors output an analog signal, which changes depending on the distance between the sensor and an object. From the datasheet, you can see that the output voltage of the SHARP GP2Y0A21YK0F ranges from 2.3 V when an object is 10 cm away to 0.4 V when an object is 80 cm away. The graph also shows why the usable detection range starts at 10 cm.

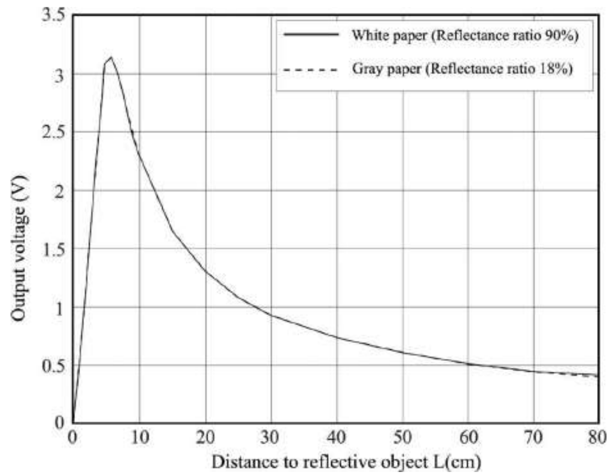


Figure 6.25: Distance measuring characteristics (output)

Table 6.5: GP2Y0A21YK0F Specifications

Operating voltage	4.5 to 5.5 V
Operating current	30 mA
Measuring range	10 to 80 cm
Output type	Analog
Dimensions	29.5 x 13 x 13.5 mm
Mounting holes	2x 3.2 mm, 37 mm spacing

6.1.2.2.2 High torque DC gear Motor with wheel

A high torque DC gear motor, often referred to as a Johnson DC gear motor, is a versatile component widely used in robotics, automation systems, and other applications requiring strong rotational force at low speeds. This motor comes with a current controlled drive for Industrial grade high torque dc motor with various types of input signals. It supports UART/I2C/PPM/Analog signals directly for speed and direction control. The drive even works well at very low speeds.

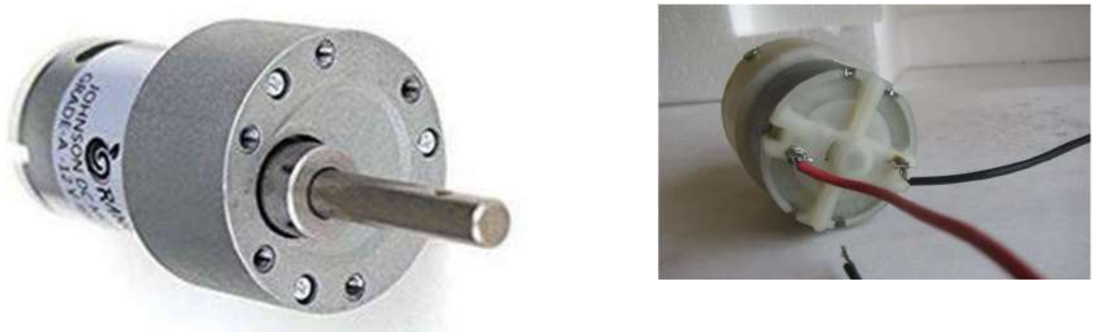


Figure 6.26: High torque DC gear Motor

The Motor is a Industrial grade 200RPM high torque motor with a massive torque of 13.5kgcm in small size. The motor has a metal gearbox with all high quality metal gears and has an off-centered shaft.

High Torque DC Geared Motor Features:

- 300RPM 12V DC motors with Metal Gearbox and Metal Gears
- 18000 RPM base motor
- 6mm Dia shaft with M3 thread hole
- Gearbox diameter 37 mm
- Motor Diameter 28.5 mm
- Length 63 mm without shaft
- Shaft length 30mm
- 180gm weight
- 13.59kgcm Holding Torque
- No-load current = 800 mA, Load current = up to 7.5 A(Max)

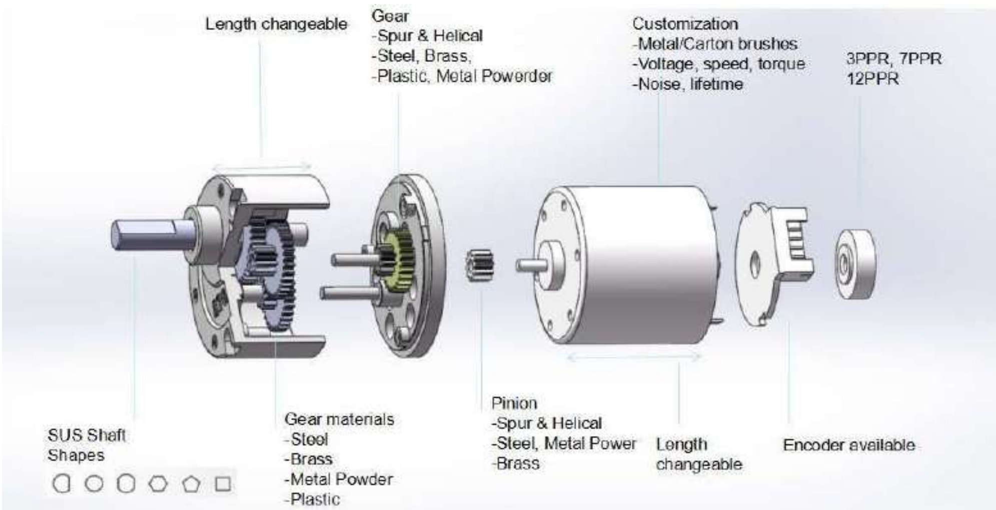


Figure 6.27: Parts of DC gear motor

Table 6.6: Power and Input Terminal Assignments:

Terminal No.	Wire Color	Terminal Name	Description
Terminal 1	BLACK	GND	Ground should be connected to negative of supply of battery
Terminal 2	BROWN	SCL/PPM/Analog	I2C clock / PPM input signal / Analog Voltage Input
Terminal 3	RED	SDA/Analog Sense	I2C Data / Analog Input Sense
Terminal 4	ORANGE	UART TXD	UART Data Transmit of speed controller, connect to RXD of host
Terminal 5	YELLOW	UART RXD	UART Data Receive of speed controller, connect to TXD of host
Terminal 6	GREEN	V+	V+ should be connected to positive of supply or battery

Current controlled Drive Specifications:

- High-Current DC Constant-Torque motor drive integrated with the motor
- Motor speed control interface via UART, I2C, PPM signal and analog input
- Speed control possible in both directions down to almost 1% of max. speed
- Small package and integration allows for easy installation and operation
- Speed can be controlled using a terminal or MCU via simple UART commands
- I2C master device can control multiple RMCS-210x via simple I2C command structures
- An RC receiver or any PPM source can directly control the speed of the motor
- An analog signal or fixed analog voltage from a potentiometer can directly control the speed of the motor

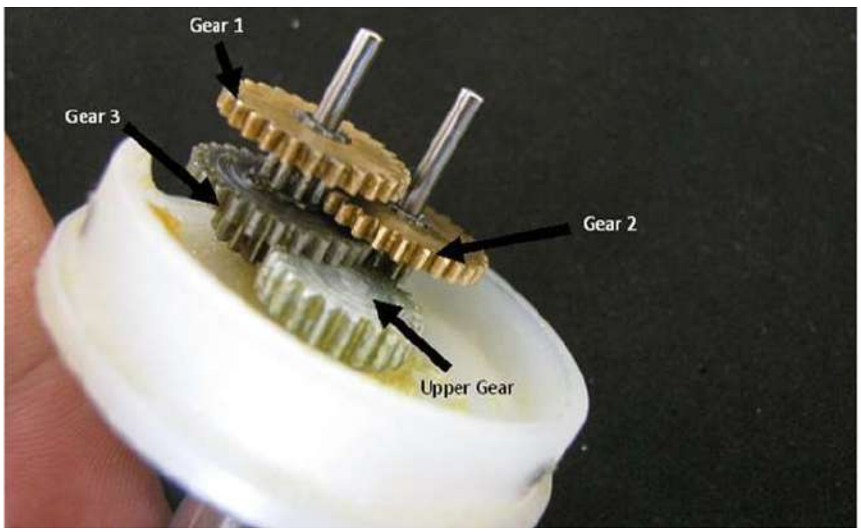


Figure 6.28: Combination of Bottom Gear Assembly

Current controlled Drive Specifications:

- High-Current DC Constant-Torque motor drive integrated with the motor
- Motor speed control interface via UART, I2C, PPM signal and analog input
- Speed control possible in both directions down to almost 1% of max. speed
- Small package and integration allows for easy installation and operation
- Speed can be controlled using a terminal or MCU via simple UART commands

- I2C master device can control multiple RMCS-210x via simple I2C command structures
- An RC receiver or any PPM source can directly control the speed of the motor
- An analog signal or fixed analog voltage from a potentiometer can directly control the speed of the motor.[19]

6.1.2.2.3 Motor Driver Shield (L298N)

L298N module is a high voltage, high current dual full-bridge motor driver module for controlling DC motor and stepper motor. It can control both the speed and rotation direction of two DC motors. This module consists of an L298 dual-channel H-Bridge motor driver IC. This module uses two techniques for the control speed and rotation direction of the DC motors. These are PWM – For controlling the speed and H-Bridge – For controlling rotation direction. These modules can control two DC motor or one stepper motor at the same time.

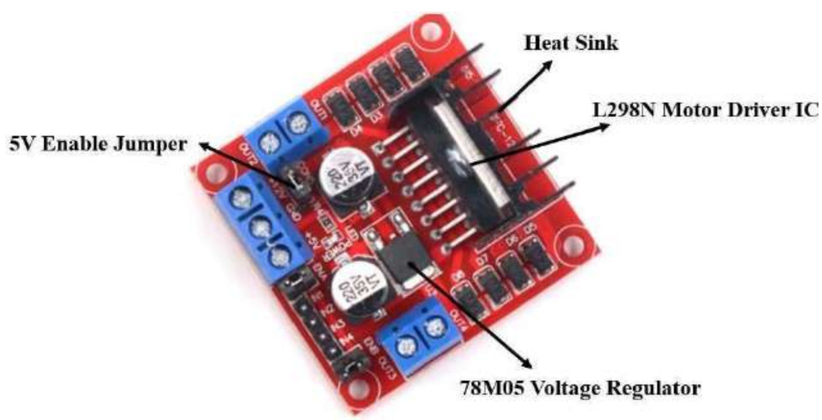


Figure 6.29:L298N Motor Driver Module

The L298N Motor driver IC is powerfully built with a big heat sink. It is a dual-channel H bridge motor driver which can be easily used to drive two motors. The module also has a 78M05 5V regulator which is enabled through a jumper. Keeping the jumper intact, means the 5V regulator is enabled.

Specifications

Table 6.7: The table shows some specifications of the L298N motor driver module:

Driver Model	L298N
Driver Chip	Double H-bridge L298N
Maximum Power	25W
Maximum Motor Supply Voltage	46V
Maximum Motor Supply Current	2A
Driver Voltage	5-35V
Driver Current	2A
Size	43x43x26mm

PinOut

Let us now look at the pin out of the module

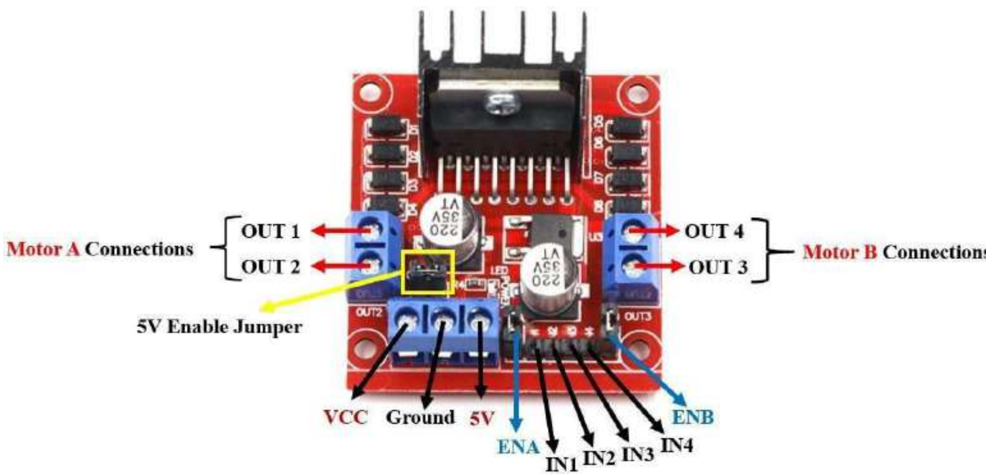


Fig6.30: the pinout of the module L289

Table 6.8: Motor driver pinout

Pin Name	**Description**
VCC	This is the pin which supplies power to the motor. It is imprinted with +12V on board but can be powered between 6-12V.
Ground	This is the common ground pin.
5V	This pin supplies the power (5V) for the internal circuit (L298N IC). Will be used only if the 5V enable jumper is not intact. If jumper is intact, then it acts as an output pin.

ENA	This pin controls the speed of the motor A by enabling the PWM signal.
IN1 & IN2	These are the input pins for motor A. They control the spinning direction for that particular motor.
IN3 & IN4	These are the input pins for motor B. They control the spinning direction for that particular motor.
ENB	This pin controls the speed of the motor B by enabling the PWM signal.
OUT1 & OUT2	OUT1: Positive terminal. OUT2: Negative terminal These are the output pins for motor A. Motor A having voltage between 5-35V, will be connected through these two terminals.
OUT3 & OUT4	OUT3: Positive terminal OUT4: Negative terminal These are the output pins for motor B.

H-Bridge Techniques

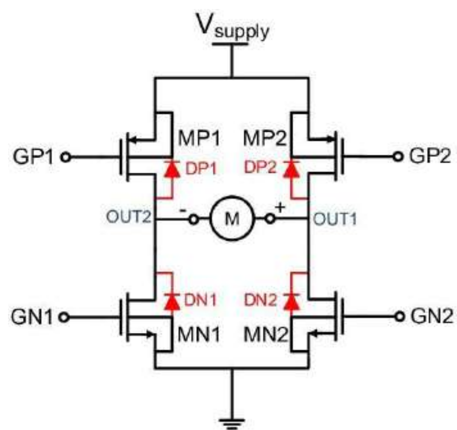


Figure 6.31: Dual H-Bridge

L298n motor driver module uses the H-Bridge technique to control the direction of rotation of a DC motor. In this technique, H-Bridge controlled DC motor rotating direction by changing the polarity of its input voltage.An H-Bridge circuit contains four switching elements, like transistors (BJT or MOSFET), with the motor at the center forming an H-like configuration.

Input IN1, IN2, IN3, and IN4 pins actually control the switches of the H-Bridge circuit inside L298N IC.We can change the direction of the current flow by activating two particular switches at the same time, this way we can change the rotation direction of the motor. [20]

6.1.2.2.4 Li-ion rechargeable battery (3.7v 2000mah 18650)

18650 battery is a Li-ion rechargeable battery with a 2000 mAh Battery Capacity. This is not a standard AA or AAA battery but is very useful for applications that require continuous high current or high current in short bursts like in cameras, DVD players, iPod, etc. They are safe to use, environment friendly and have long battery life. This li ion battery comes with high energy density and provides excellent continuous power sources to your device. It should be used with a protection circuit board that guards the battery against over-charge, over-discharge of the pack, and avoid over-current drawn.

Figure 6.32: Li-ion rechargeable battery (3.7v 2000mah 18650)

Table 6.9 battery Specifications:

Voltage	3.7V
Battery Capacity	2000 mAh
Battery Type	Lithium-Ion
Charging method	CC-CV

Rechargeable	Yes
Dimensions (LxD)	67 x 18mm
Weight	30 grams
Maximum charging voltage	4.2V
Maximum charging current	1000mA

6.1.2.3 Connection Diagram:

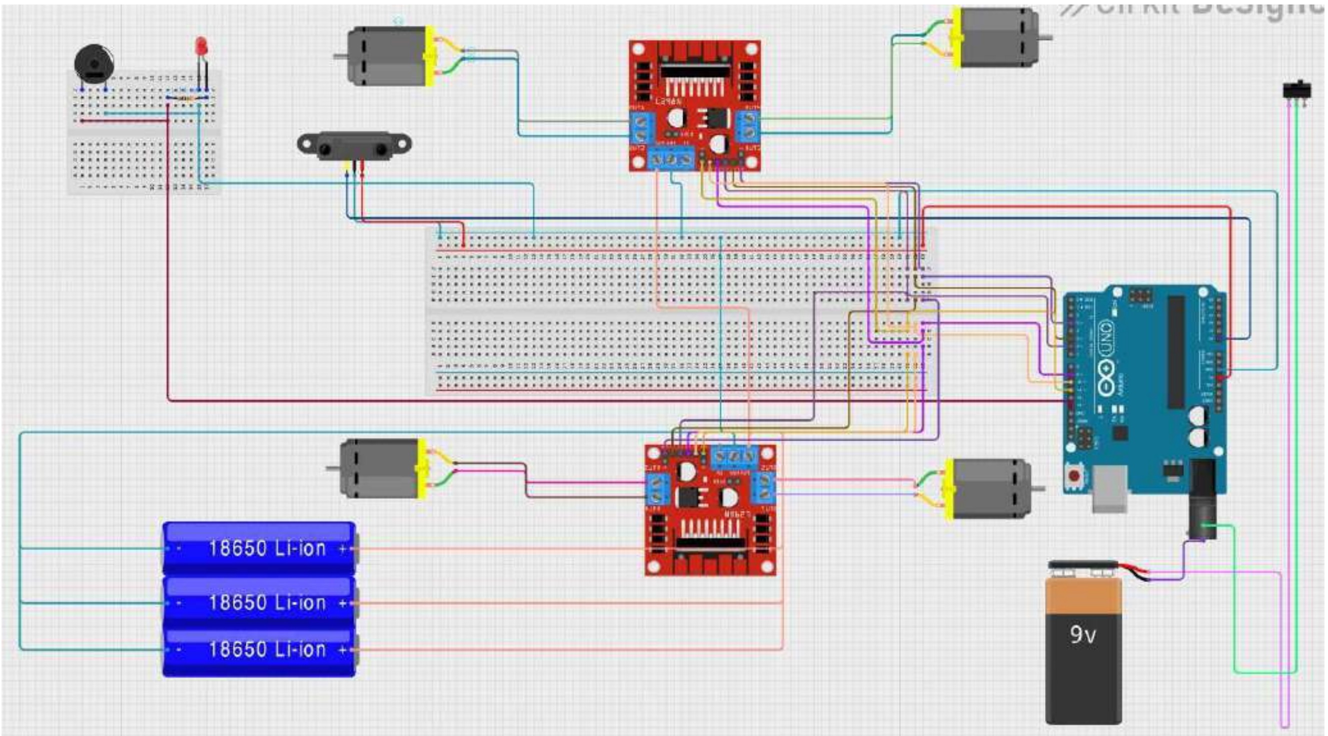


Figure 6.33: The connection diagram of Locomotive sensing and Automatic breaking system

6.1.2.4 IN MODEL

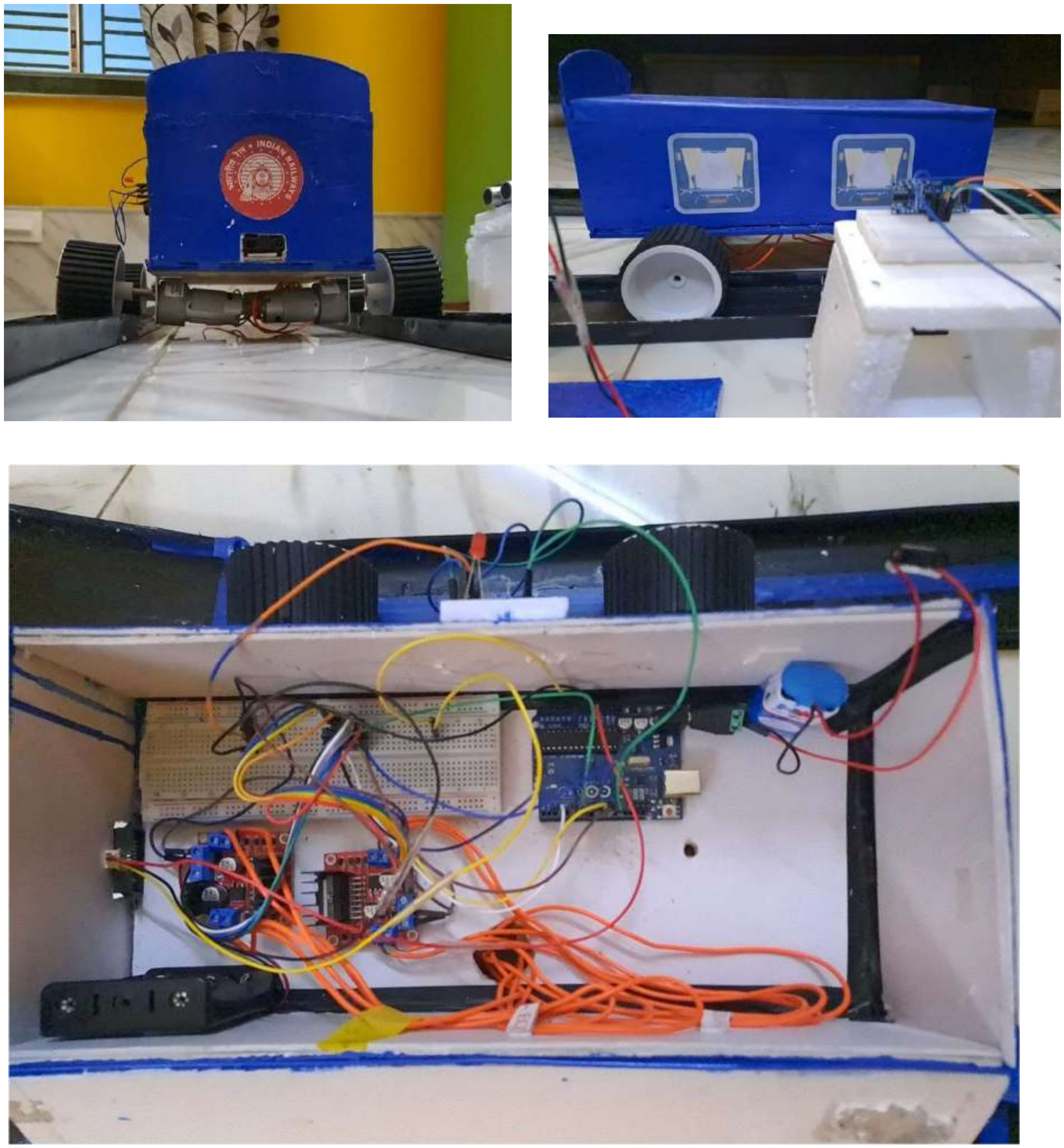


Figure 6.34: In model picture of Locomotive sensing and Automatic breaking system

6.1.2.5 FLOWCHART

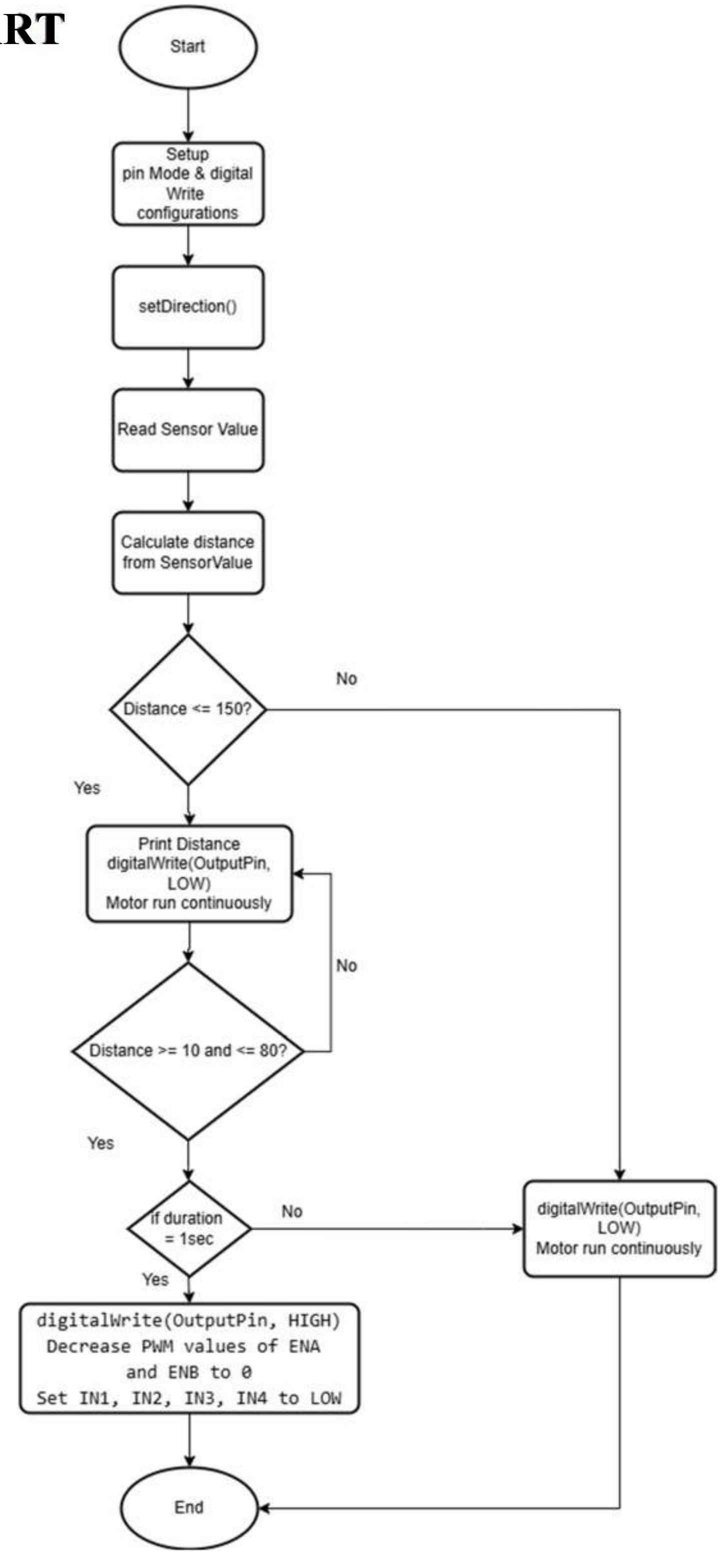


Figure 6.35: Flowchart for Locomotive sensing and Automatic breaking system

6.1.2.6 CODE FOR LOCOMOTIVE SENSING AND AUTOMATIC BREAKING SYSTEM

```
#define sensor A0
int ENA = 11;
int IN1 = 10;
int IN2 = 9;
int IN3 = 6;
int IN4 = 5;
int ENB = 3;
int OutputPin = 13;

unsigned long startTime = 0;
bool objectDetected = false;

void setup() {
  pinMode(ENA, OUTPUT);
  pinMode(ENB, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);

  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
  pinMode(OutputPin, OUTPUT);
  pinMode(sensor, INPUT);
  Serial.begin(9600);
}

void loop() {
  setDirection();
  int SensorValue = analogRead(sensor);
```

```
 float volts = SensorValue * 0.0048828125;
 int distance = 29.988 * pow(volts, -1.173);
 delay(100);

 if (distance <= 150) {
   Serial.println(distance);
 }

 if (distance >= 10 && distance <= 80) {
   if (!objectDetected) {
     objectDetected = true;
     startTime = millis();
   } else if (millis() - startTime >= 1000) { // Object has been in range
for at least 1 second
     digitalWrite(OutputPin, HIGH);
     for (int pwmValue = 50; pwmValue >= 0; pwmValue--) {
       analogWrite(ENA, pwmValue);
       analogWrite(ENB, pwmValue);
       delay(10); // delay time as needed for desired stopping time
     }
     analogWrite(ENA, 0);
     analogWrite(ENB, 0);
     digitalWrite(IN1, LOW);
     digitalWrite(IN2, LOW);
     digitalWrite(IN3, LOW);
     digitalWrite(IN4, LOW);
     delay(50000);
   }
 } else {
   objectDetected = false;
   digitalWrite(OutputPin, LOW);
 }
}

void setDirection() {
 analogWrite(ENA, 50);
```

```
  analogWrite(ENB, 50);

  digitalWrite(IN1,  LOW);
  digitalWrite(IN2, HIGH);
  digitalWrite(IN3,  LOW);
  digitalWrite(IN4, HIGH);
}
```

6.1.3 Accident prevention signal

6.1.3.1 APPROACH

In our model, we have used an Accident prevention signal with **two Ultrasonic Sensors** in the track. This model is designed to detect the presence of any other locomotive in the same track. It also measure the speed of a locomotive passing between two ultrasonic sensors using an Arduino. The system comprises two **ultrasonic sensors and an LED indicators**. The LED will give the signal to the locomotive if there is any other locomotive coming in the same track. Here's a detailed theoretical approach to understand how this system works:

In the loop function, the system continuously monitors the sensors and performs the following steps:

Triggering and Reading Ultrasonic Sensors:

For each sensor, a pulse is sent from the trigger pin, and the echo pin listens for the pulse to return after bouncing off an object.

The pulseIn function measures the duration of the returning pulse. The distance to the object is calculated using the formula:

$$\textbf{Distance} = (\textbf{duration} \times 0.034) / 2$$

This formula is based on the speed of sound in air (0.034 cm/μs).

Object Detection:

If the distance measured by the first sensor (distance1) is less than or equal to 10 cm, it records the current time using millis() and sets flag1.

Similarly, if the distance measured by the second sensor (distance2) is less than or equal to 10 cm, it records the current time and sets flag2.

Calculating Speed:

When both sensors have detected the object (flag1 and flag2 are both set), the time difference between the detections (Time) is calculated.

The speed is then calculated using the formula:

$$\textbf{Speed} = \textbf{distanceBetweenSensors} / \textbf{Time}$$

The speed is converted from meters per second to kilometers per hour.

Speed Indication:
The speed is printed to the Serial Monitor.
If the speed exceeds 0.5km/h, the LED is turned on, indicating that a locomotive is detected in the same track. Otherwise, the LED remains off.
The system then waits for 30 seconds, resets the flags, and the LED is turned off.

Idle and Searching States:
If no locomotive is detected by both sensors, a "No Locomotive detected" message is printed, and the system waits for 1 second before checking again.
If one sensor has been triggered but not the other, it prints "Searching..." indicating that it is waiting for the second sensor to detect the object.

6.1.3.2 COMPONENTS USED

6.1.3.2.1 Ultrasonic sensor:

An ultrasonic sensor is an instrument or an electronic device that measures the distance to target object using ultrasonic sound waves. It ultrasonic sensor uses a transducer for sending and receiving the ultrasonic pulses which delivers us information about proximity of an object. An ultrasonic sensor transforms the sound reflected to an electric signal. Ultrasonic waves would travel faster compared to the speed of sound which is audible to humans.

Figure 6.36: Ultrasonic distance sensor

The ultrasonic sensor has two components namely:

10. Transmitter
11. Receiver

Transmitter uses piezoelectric crystals, the transmitter emits the sound. **Receiver** finds the sound after it is traveled to and from the target.

Table 6.10: Four pins of the ultrasonic sensor:

Vcc	Power supply +5 V
Gnd	Common ground
Trigger pin	To start the sensor
Eco pin	Receive the signal

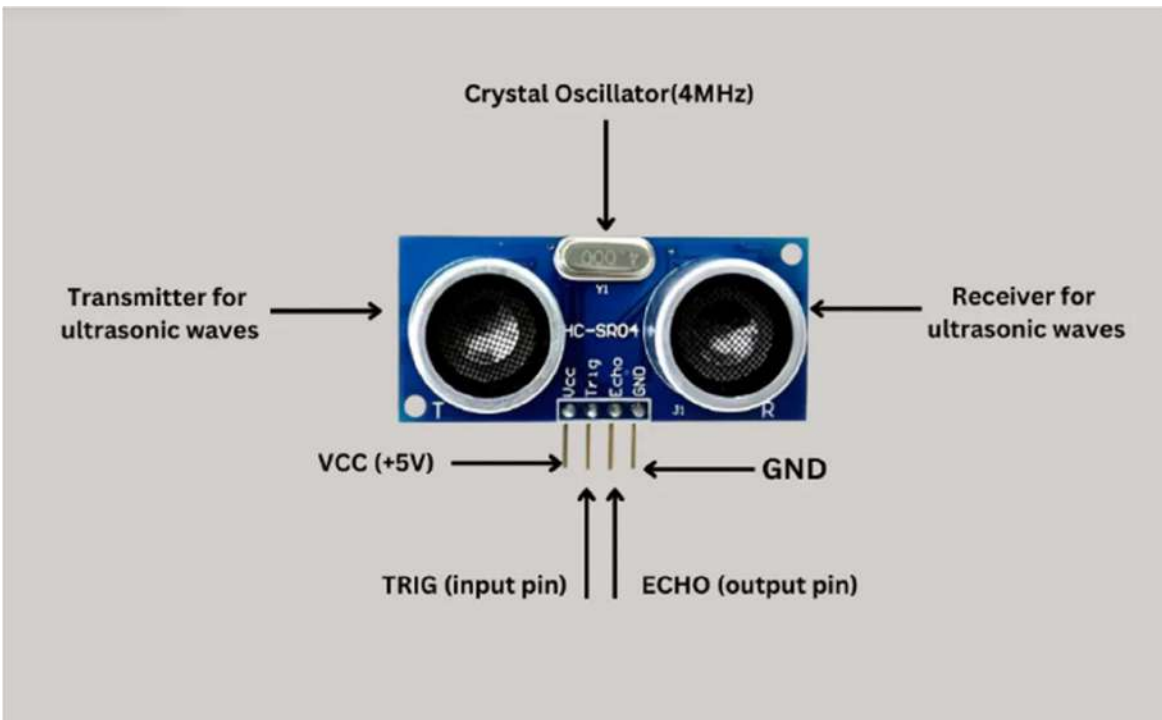


Figure 6.37: Pins of ultrasonic sensor

The principle of the ultrasonic rangefinders is to measure the time during which that signal propagates at the distance from the transmitter to the receiver. The propagation speed of the signal is known. The sensor consists of a transmitter that generates ultrasonic waves, a receiver that perceives the echo, and auxiliary nodes for normal operation of the module.

The ultrasonic rangefinder HC-SR04 is shown in Figure, where **the transmitter and the receiver** are designated as T and R, respectively. The rangefinder generates sound waves at a frequency of 40 kHz. The sound waves reflects from the object and returns to the receiver, the sensor gives the information about the time, which was demanded to sound waves for propagation from the sensor to the object and back. This process can be clearly seen in Figure.[25]

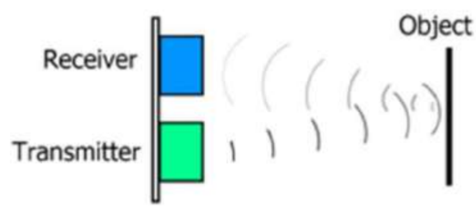


Figure 6.38: Ultrasound rangefinder

Figure 6.39: Movement of the ultrasonic

The ultrasonic signal is propagated by a wave directed at an angle of 30°. The direction of propagation of the ultrasonic signal from the transmitter is shown in Figure. The most effective measuring angle is 15°. External objects falling under this measurement angle interfere with the determination of the distance to thedesired object.

The readings of the **ultrasound range finders are not affected by sunlight or the color of objects**, as is the case with infrared sensors. The ultrasonic wave is reflected from almost any surface, even transparent, but it can be difficult to determine the distance to fluffy or small objects. In addition, the readings are affected by the angle of incidence of the wave.

If the sensor is directed perpendicular to the object, the measurements will be most accurate. In addition, if, the angle of incidence is too large, then the wave, reflected from the object, does not enter the receiver, which will lead to an incorrect measurement.

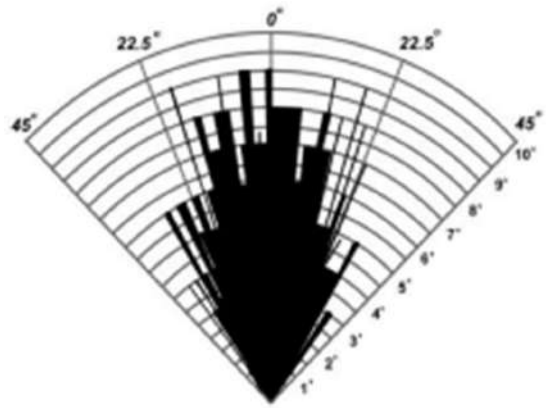


Figure 6.40: The ultrasonic wave pattern

For **calculating the distance** between the sensor and object, sensor measures the time it takes for the transmitter to emit the sound and to its contact with receiver.**D = (T*C)/2 [**where D = distance, T = time, C = speed of the sound]
Ultrasonic sensors are used mainly as proximity sensors. Ultrasonic sensors are primarily found in anti-collision safety systems, automobile self-parking technology, robotic obstacle detection system and manufacturing system.
With microcontroller platforms like the Raspberry Pi, ARM, PIC, **Arduino IDE**, Beagle Board, and many more, our ultrasonic proximity level and distance sensors are often employed.
Ultrasonic sensors will send sound waves in the direction of the target and calculate its distance by timing how long it takes for the waves to bounce back to the sensor. In addition, to collision avoidance systems also employ ultrasonic sensors

6.1.3.3 CONNECTION DIAGRAM

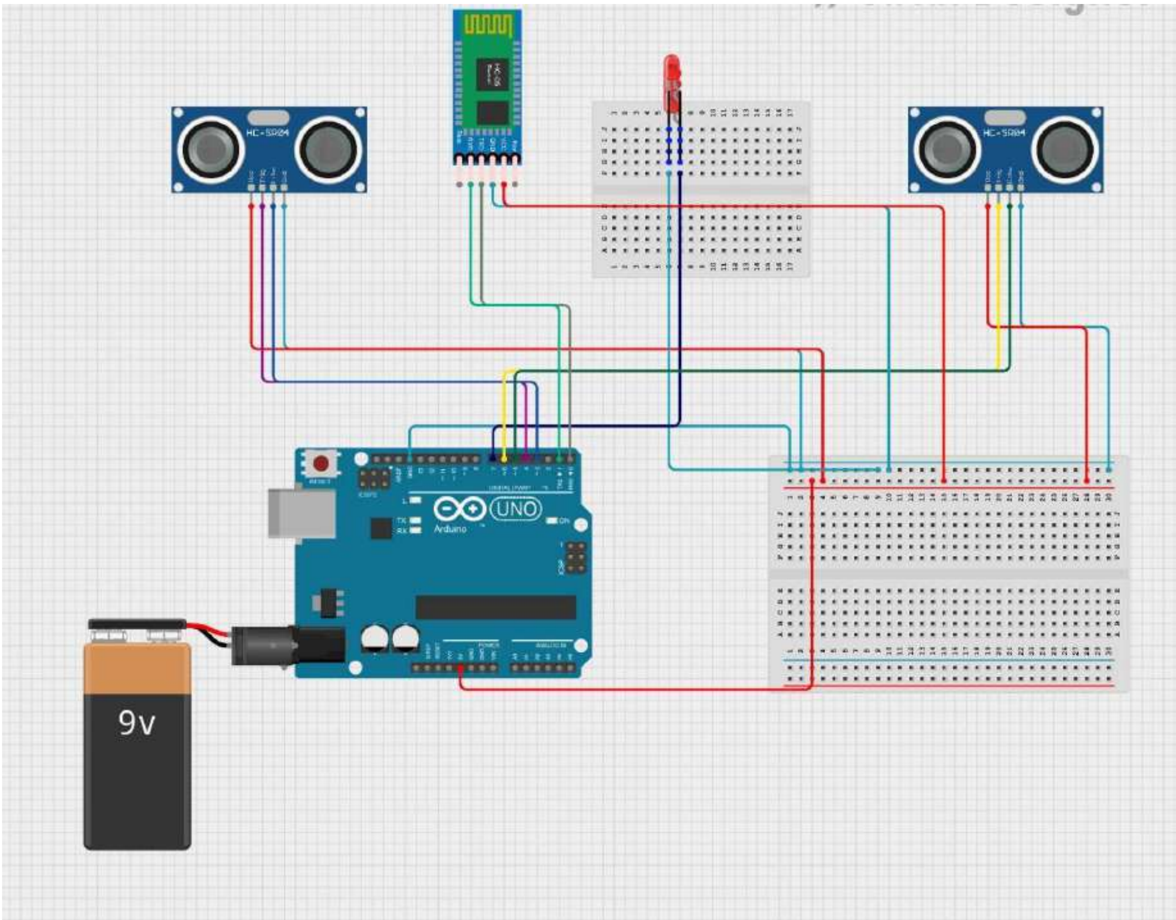


Figure 6.41: Connection diagram for Accident prevention signal

We have used a software called Circuit Designer for the development of this connection diagram.

6.1.3.4 IN MODEL

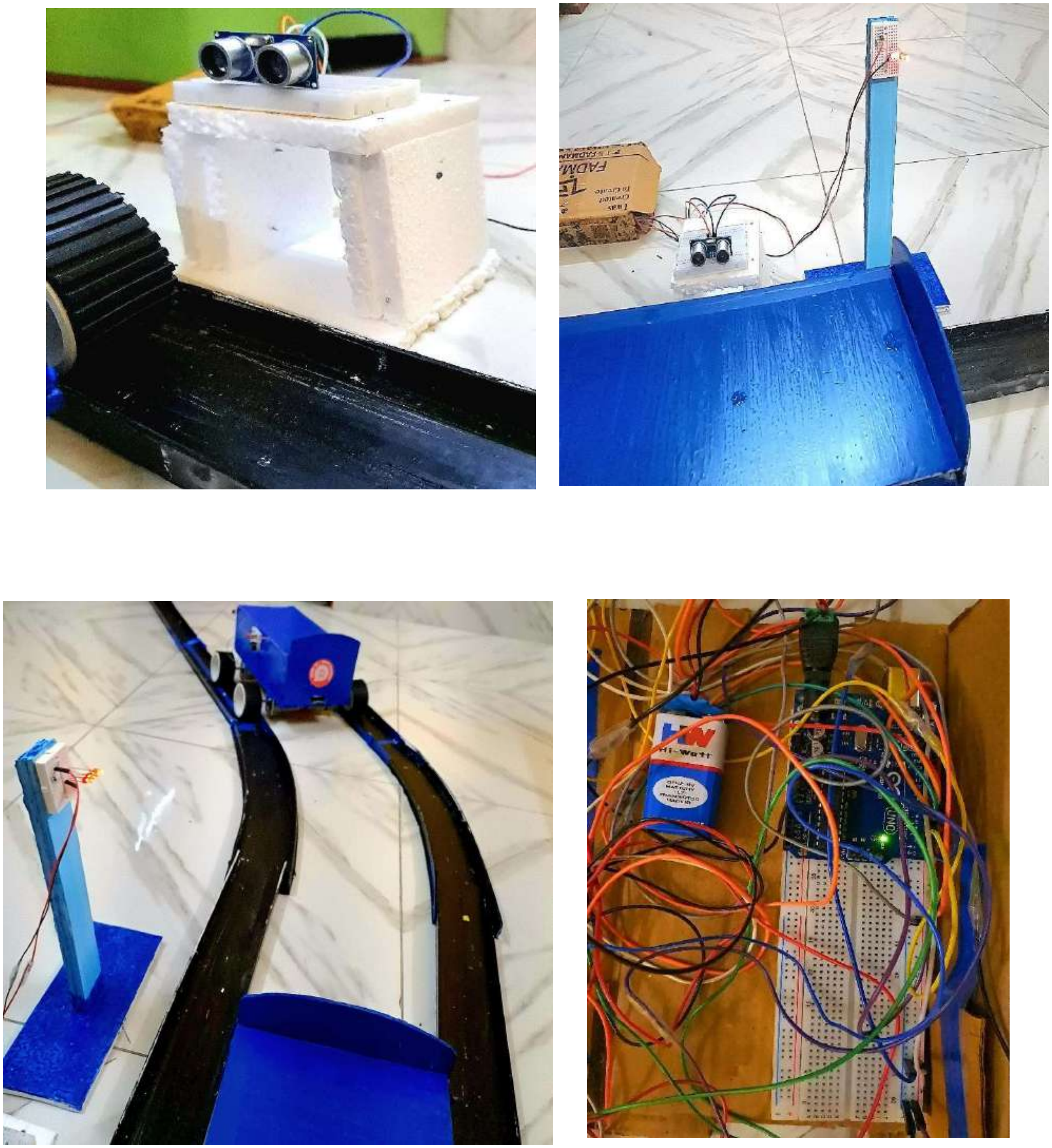


Figure 6.42: In-Model Picture of Accident prevention signal

6.1.3.5 FLOWCHART

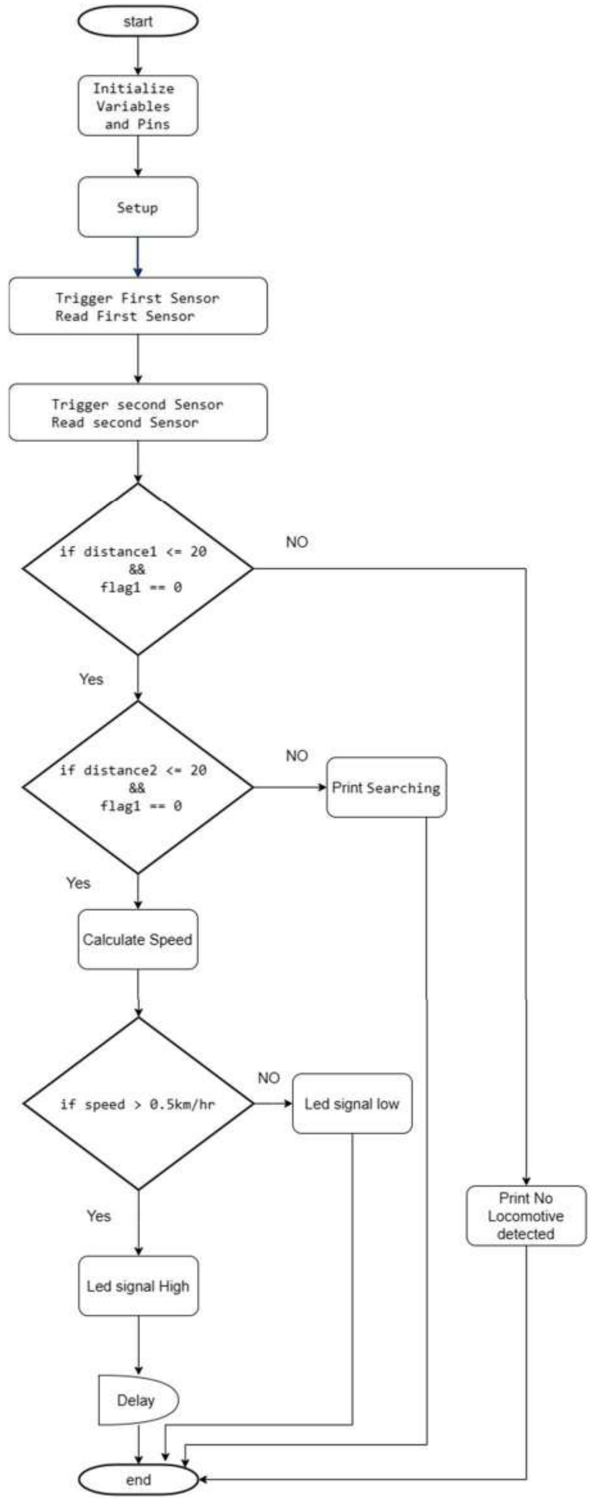


Figure 6.43: Flowchart of Accident prevention signal

6.1.3.6 CODE FOR ACCIDENT PREVENTION SIGNAL

```
int timer1;
int timer2;
float Time;
int flag1 = 0;
int flag2 = 0;
float distanceBetweenSensors = 0.60; // Distance between sensors in meters
float speed;

int trig1 = 4;
int echo1 = 3;
int trig2 = 6;
int echo2 = 5;
int led = 7;

void setup() {
  pinMode(trig1, OUTPUT);
  pinMode(echo1, INPUT);
  pinMode(trig2, OUTPUT);
  pinMode(echo2, INPUT);
  pinMode(led, OUTPUT);
  Serial.begin(9600); // Initialize serial communication for debugging
}

void loop() {
  long duration1, duration2;
  float distance1, distance2;

  // Trigger the first ultrasonic sensor
  digitalWrite(trig1, LOW);
  delayMicroseconds(2);
  digitalWrite(trig1, HIGH);
  delayMicroseconds(10);
  digitalWrite(trig1, LOW);

  // Read the first ultrasonic sensor
  duration1 = pulseIn(echo1, HIGH);
  distance1 = (duration1 * 0.034 / 2);

  // Trigger the second ultrasonic sensor
  digitalWrite(trig2, LOW);
  delayMicroseconds(2);
  digitalWrite(trig2, HIGH);
  delayMicroseconds(10);
  digitalWrite(trig2, LOW);
// Read the second ultrasonic sensor
  duration2 = pulseIn(echo2, HIGH);
  distance2 = (duration2 * 0.034 / 2);
```

```
  // Check if the object is detected by the first sensor within 20 cm
  if (distance1 <= 20 && flag1 == 0) {
    timer1 = millis();
    flag1 = 1;
    Serial.println("Sensor 1 triggered");
  }

  // Check if the object is detected by the second sensor within 20 cm
  if (distance2 <= 20 && flag2 == 0) {
    timer2 = millis();
    flag2 = 1;
    Serial.println("Sensor 2 triggered");
  }

  if (flag1 == 1 && flag2 == 1) {
    if (timer1 > timer2) {
      Time = timer1 - timer2;
    } else if (timer2 > timer1) {
      Time = timer2 - timer1;
    }
    Time = Time / 1000; // Convert milliseconds to seconds
    speed = distanceBetweenSensors / Time; // v = d/t
    speed = speed * 3600; // Convert to km/h
    speed = speed / 1000; // Convert to km/h

    // Print speed to the Serial Monitor
    Serial.print("Speed: ");
    Serial.print(speed, 1);
    Serial.println(" Km/Hr");

    if (speed > 0.5) {
      Serial.println("Locomotive detected");
      digitalWrite(led, HIGH);
    } else {
      Serial.println("Very low speed object");
    }
     delay(30000);
    digitalWrite(led, LOW);
    speed = 0;
    flag1 = 0;
    flag2 = 0;
  } else {
    if (flag1 == 0 && flag2 == 0) {
      Serial.println("No Locomotive detected");
      delay(1000);
    } else {
      Serial.println("Searching...");
    }
  }
  delay(10); // Small delay to prevent overwhelming the serial output
}
```

6.1.4 Accident alert system

6.1.4.1 APPROACH

The primary objective of this model is to use an **MPU6050 accelerometer and gyroscope sensor** to detect abnormal motion (such as **an accident**) and trigger an alert using a buzzer or led and sent alert to the **nearest control room for emergency** when any kind of accident occurred.

This model continuously monitors the accelerometer values from the MPU6050 sensor. If the values indicate that the device has **experienced motion outside of normal parameters** (indicating a possible accident), the alert is activated. The sensor values are read and processed every second, ensuring timely detection of any abnormal motion.

This is designed to detect significant changes in motion using an MPU6050 sensor. When such changes are detected, it triggers an alert. The main components of the system include data acquisition from the sensor, data processing and mapping, condition checking against predefined thresholds, and triggering an alert when abnormal motion is detected. **The system runs continuously, rechecking the conditions every second**. This makes it suitable for applications where continuous monitoring for sudden changes in motion is required. In the subsequent sections, we will look into the theoretical steps involved in conceptualizing, implementing, and deploying the model using the MPU6050 sensor.

Sensor Introduction:

Understand the purpose of the sensor: The **MPU6050** is a 6-axis motion- tracking device that combines a 3-axis gyroscope and a 3-axis accelerometer ina single chip.

Sensor Installation:

Physically connect the sensor to the microcontroller (in this case, an Arduino board).

Ensure proper wiring and connections according to the sensor's pinout diagram.

Data Processing:

Normalize the raw sensor data to make it more usable and interpretable.

Map the raw sensor data to a desired range using the map() function.

Threshold Setting:
Determine threshold values for detecting abnormal motion based on the application requirements.
Establish threshold ranges for each axis (X, Y, Z) of the accelerometer to indicate abnormal motion.

Abnormal Motion Detection:
Compare the normalized sensor data against the predefined thresholds.
Check if the sensor readings fall outside the expected ranges for each axis.

Alert Mechanism:
Activate an alert mechanism (e.g., a buzzer) if abnormal motion is detected.
Optionally, trigger additional actions or notifications based on the detected motion.

Error Handling:
Implement error-handling mechanisms to address potential issues such as sensor disconnection or malfunction.
Include safeguards to ensure the reliability and robustness of the sensor system.

Testing and Calibration:
Test the sensor system under different conditions to ensure its accuracy and reliability.
Calibrate sensor settings if necessary to optimize performance and minimize false positives or false negatives.

Integration and Deployment:
Integrate the sensor system into the target application or device.
Deploy the sensor system in real-world scenarios and monitor its performance over time.

Continuous Monitoring and Maintenance:
Regularly monitor the sensor system to ensure its continued functionality and accuracy.
Perform routine maintenance tasks such as cleaning and recalibration as needed.
Address any issues or anomalies promptly to maintain the effectiveness of the sensor system.

6.1.4.2 COMPONENTS USED

6.1.4.2.1 Gyroscope sensor:

A **gyroscope sensor** is basically a device that takes the help of the earth's gravity in determining the orientation. It is a type of sensor which we find inside IMU (Inertial Measurement Unit). A gyroscope can be used to measure the rotation on a **particular axis**. The device consists of a rotor which is nothing but a freely rotating disc.

Gyroscope sensor is a device that can measure and maintain the orientation and angular velocity of an object. These are more advanced than accelerometers. These can measure **the tilt and lateral orientation of the object** whereas accelerometer can only measure the linear motion.

A gyroscope sensor is a device used to **sense and maintain direction**, designed based on the theory of indestructible angular momentum. Once the gyroscope starts to rotate, due to the angular momentum of the wheel, the gyroscope has a tendency to resist changes in direction. The principle of a gyroscope is that the **direction pointed by the axis of rotation** of a rotating object will not change when it is not affected by external forces. People use it to keep their direction based on this principle. Then use a variety of methods to read the **direction indicated by the axis, and automatically transmit the data signal to the control system.**

A modern gyroscope sensor can accurately determine the position of moving objects. It is an inertial navigation instrument widely used in modern aviation, navigation, aerospace, and defense industries. The main part of the traditional inertial gyroscope is a mechanical gyroscope, and the mechanical gyroscope has **high requirements on the process structure**. The basic idea of modern fiber-optic gyroscopes was put forward in the **1970s**. After the 1980s, fiber optic gyroscopes have developed very rapidly, and laser resonant gyroscopes have also made great progress. [34]

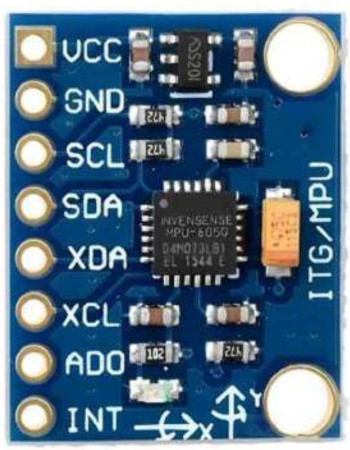


Figure 6.44: MPU6050 Gyroscope and Accelerometer Sensor

A gyroscope sensor works on the principle of **conservation of angular momentum.** It works by preserving the **angular momentum**. In a gyroscope sensor, a rotor or a spinning wheel is mounted on a pivot. The pivot allows the rotation of the rotor on a **particular axis which is called a gimbal**. Here, we will use two gimbals at a time. One gimbal will be mounted on another. This will givethe rotor three degrees of freedom. Whenever we spin the rotor of the gyroscope, the gyroscope will continue to point in the same direction.

Let us further explain the working of Vibration Gyroscopic Sensor. The Vibration Gyroscope sensor consists of a **double-T structure element**. There is a drive arm that rotates in a particular direction. The drive arms are attached to the sensing arms.When the Vibration Gyroscopic Sensor is rotated, the Coriolis Effect or the Coriolis force will start to work on the drive arms. The vibration of the drive arms causes motion in the pair of sensing arms of the gyro which will generate a potential different from the angular velocity which has been sensed by the device.

Some of the most important applications of the gyroscope sensor are:

Sensing Angular Velocity: It can be used to sense the rate of change of angular motion in moving bodies. This can be used for detecting athletic movement.

Sensing Angles: The angles can also be detected using the gyroscope sensor. This application is used in car navigations and game controllers.

Sensing Control Mechanism: We can also use a gyroscopic sensor to detect vibration due to various external factors. We can use this application for camera-shake control and vehicle control.[41]

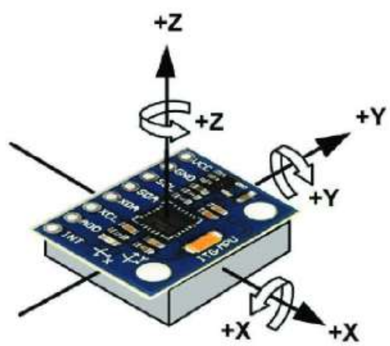


Figure 6.45: X, Y, Z axis of Gyroscope Sensor

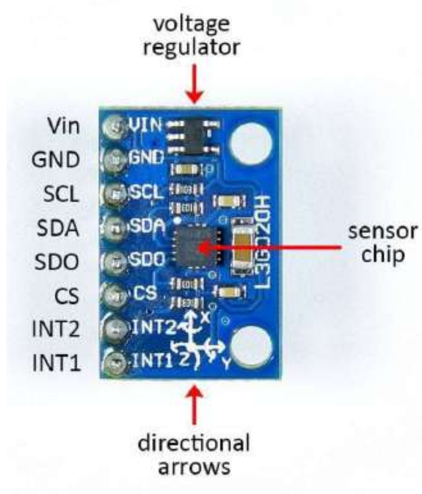


Figure 6.46: Pin diagram of Gyroscope sensor

6.1.4.2.2 HC-05 Bluetooth module:

Wireless communication is swiftly replacing the wired connection when it comes to electronics and communication. Designed to replace cable connections **HC-05 uses serial communication** to communicate with the electronics. Usually, it is used to connect small devices like mobile phones using a short-range wireless connectionto exchange files. It uses the 2.45GHz frequency band. The transfer rate of the data can vary up to 1Mbps and is in range of 10 meters.

The HC-05 module can be operated within **4-6V of power supply**. **It supports baud rate of 9600, 19200, 38400, 57600, etc.** Most importantly it can be operated in Master-Slave mode which means it will either send or receive data from external sources.[46]

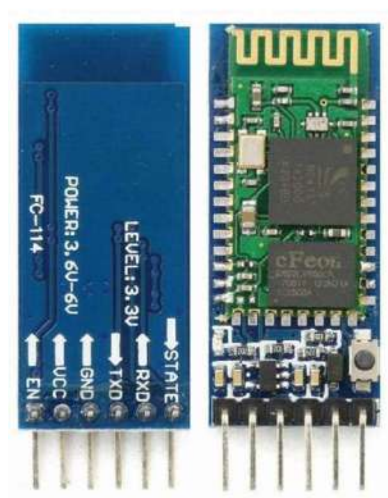


Figure 6.47: HC-05 Bluetooth module

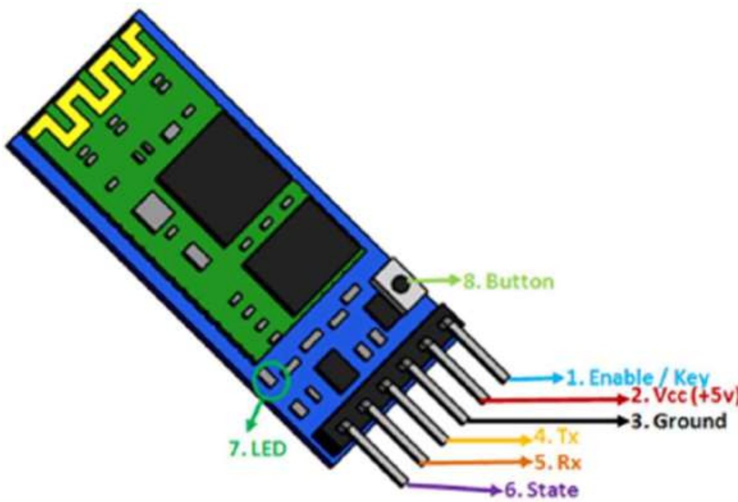


Figure 6.48: HC-05 Pinout Configuration

Table 6.11: HC-05 pinout:

Pin Number	Pin Name	Description
1	Enable / Key	This pin is used to toggle between Data Mode (set low) and AT command mode (set high). By default it is in Data mode
2	Vcc	Powers the module. Connect to +5V Supply voltage
3	Ground	Ground pin of module, connect to system ground.
4	TX – Transmitter	Transmits Serial Data. Everything received via Bluetooth will be given out by this pin as serial data.
5	RX – Receiver	Receive Serial Data. Every serial data given to this pin will be broadcasted via Bluetooth

6	State	The state pin is connected to on board LED, it can be used as a feedback to check if Bluetooth is working properly.
7	LED	Indicates the status of Module 1. Blink once in 2 sec: Module has entered Command Mode 2. Repeated Blinking: Waiting for connection in Data Mode 3. Blink twice in 1 sec: Connection successful in Data Mode
8	Button	Used to control the Key/Enable pin to toggle between Data and command Mode

HC-05 Default Settings:

4. Default Bluetooth Name: “HC-05”
5. Default Password: 1234 or 0000
6. Default Communication: Slave
7. Default Mode: Data Mode
8. Data Mode Baud Rate: 9600, 8, N, 1
9. Command Mode Baud Rate: 38400, 8, N, 1
10. Default firmware: LINVOR

HC-05 Technical Specifications:

- Serial Bluetooth module for Arduino and other microcontrollers
- Operating Voltage: 4V to 6V (Typically +5V)
- Operating Current: 30mA
- Range: <100m
- Works with Serial communication (USART) and TTL compatible
- Follows IEEE 802.15.1 standardized protocol
- Uses Frequency-Hopping Spread spectrum (FHSS)
- Can operate in Master, Slave or Master/Slave mode
- Can be easily interfaced with Laptop or Mobile phones with Bluetooth
- Supported baud rate: 9600,19200,38400,57600,115200,230400,460800.[46]

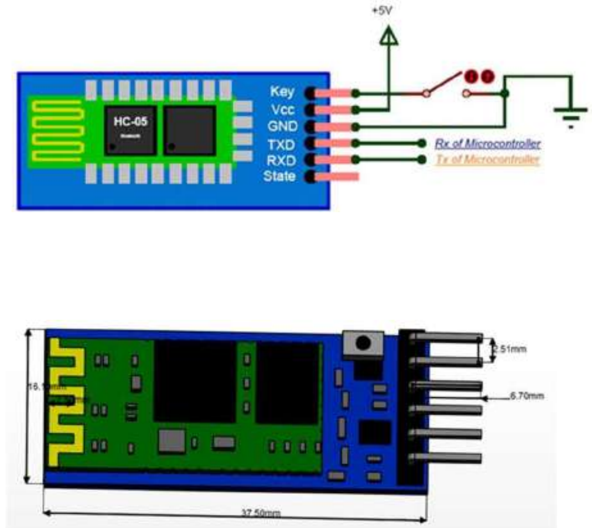


Figure 6.49: 2D model of HC-05 Bluetooth Module

6.1.4.3 CONNECTION DIAGRAM

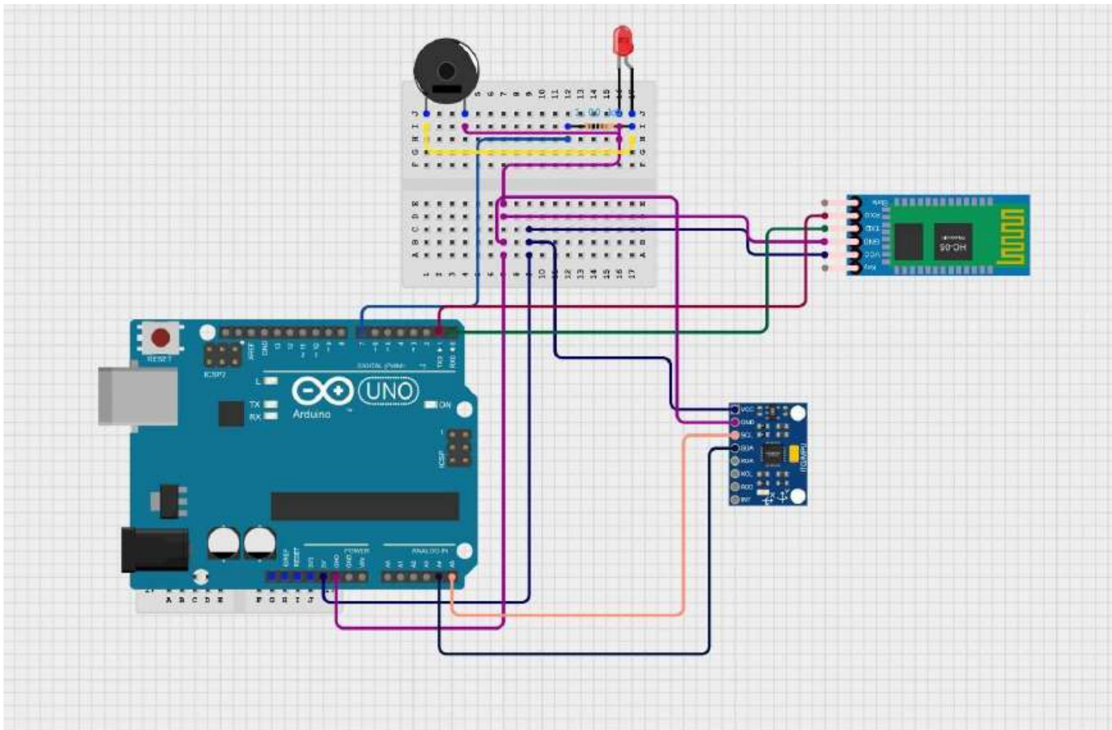
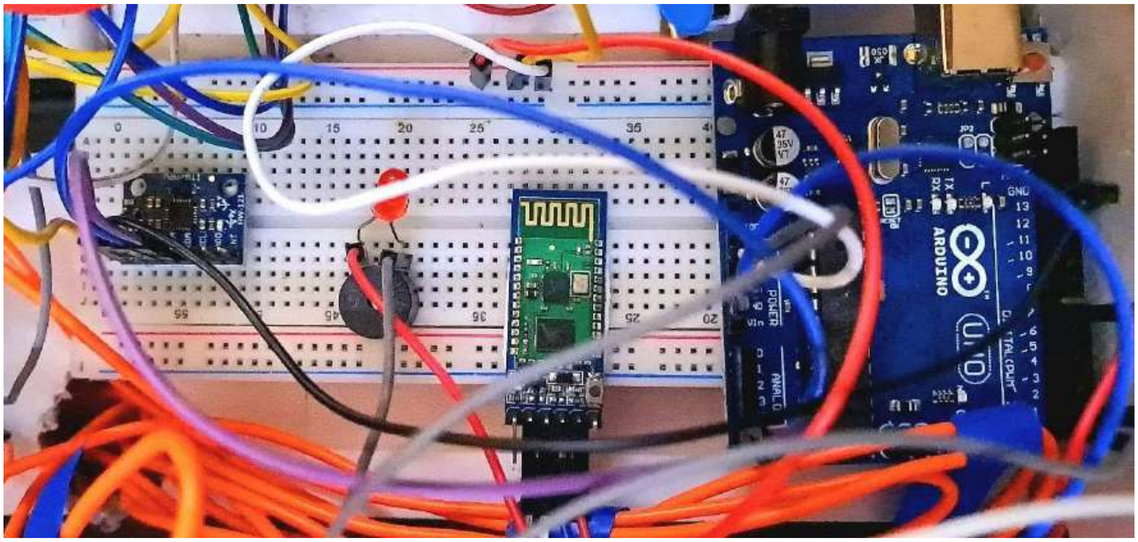


Figure 6.50: Circuit connection diagram for Accident alert system

6.1.4.4 IN MODEL



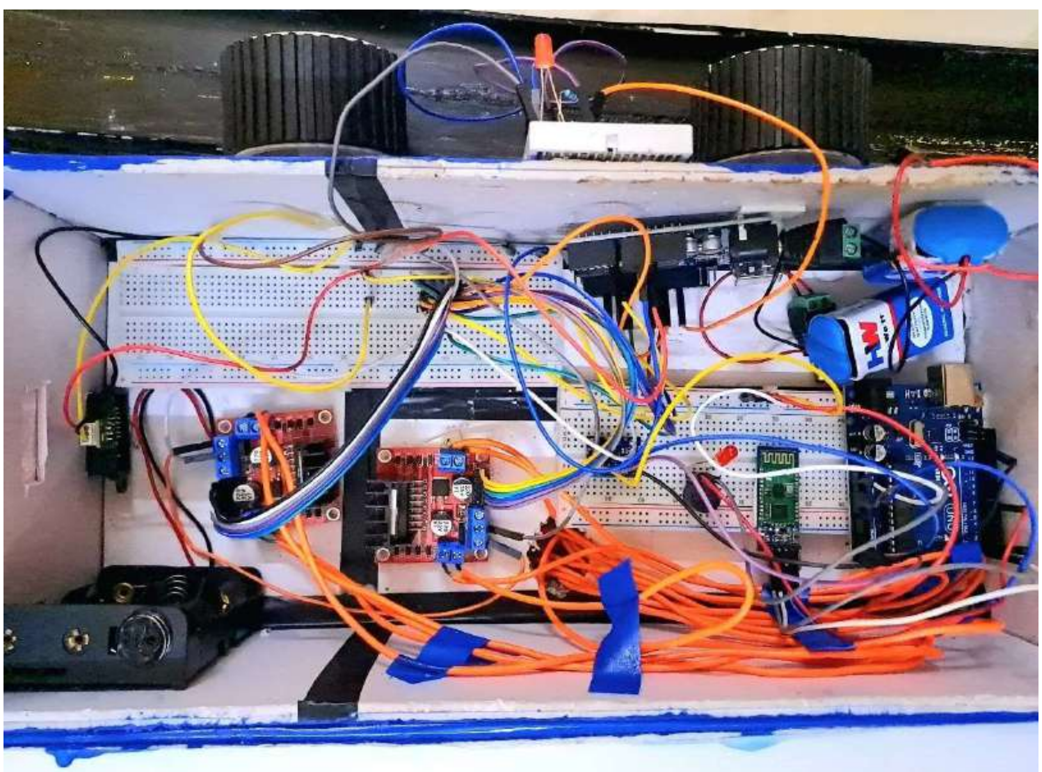


Figure 6.51: In-Model Picture of Accident alert system

6.1.4.5 FLOWCHART

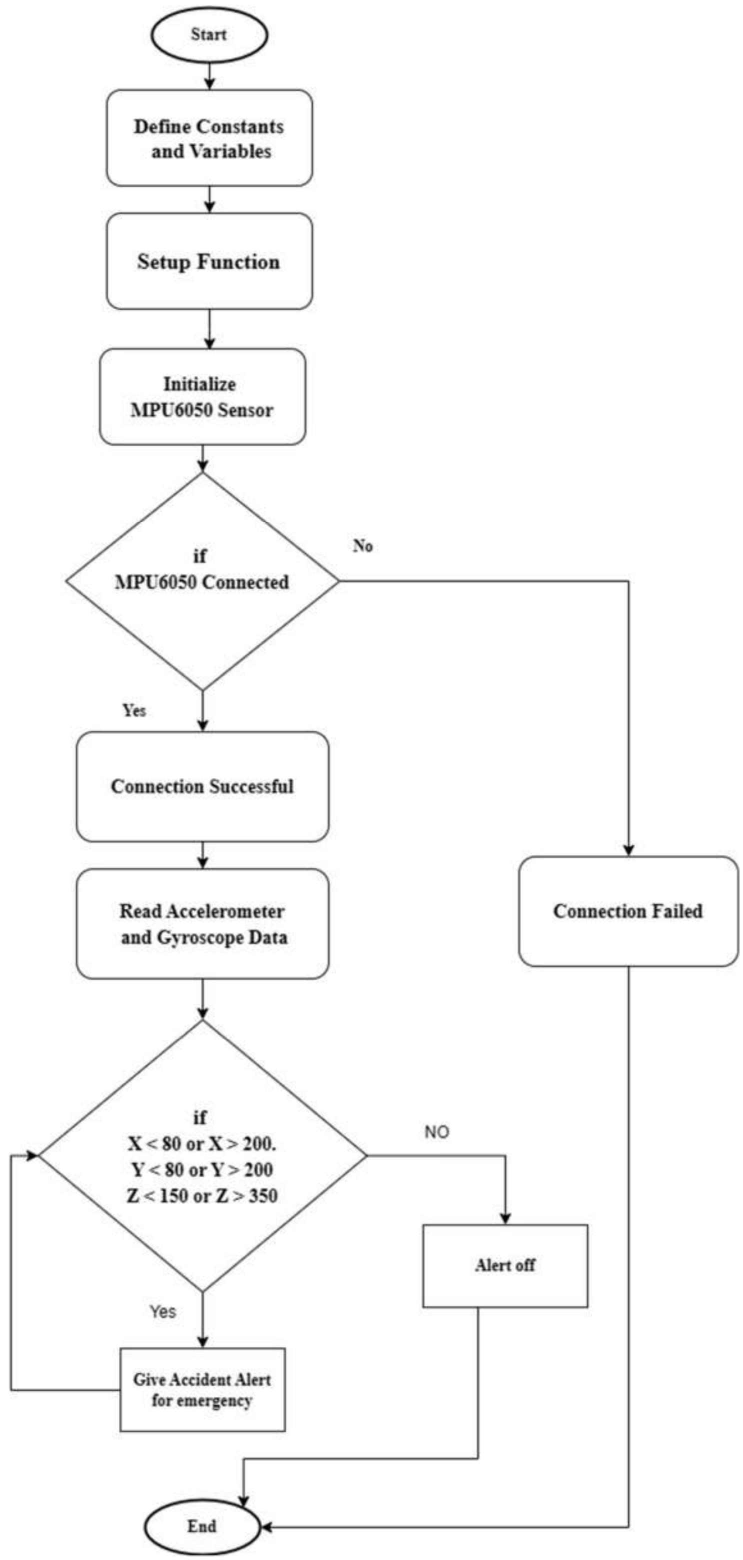


Figure 6.52: Flowchart for Accident alert system

6.1.4.6 CODE FOR Accident alert system

```
#include <Wire.h>
#include <I2Cdev.h>
#include <MPU6050.h>

#define LED_PIN 7 // Digital pin connected to the buzzer

MPU6050 mpu;
int16_t ax, ay, az;
int16_t gx, gy, gz;

struct MyData {
  byte X;
  byte Y;
  byte Z;
};

MyData data;
void setup() {
  Serial.begin(9600);
  Wire.begin();
  mpu.initialize();

  // Check if MPU6050 is connected
  if (mpu.testConnection()) {
    Serial.println("MPU6050 connection successful");
  } else {
    Serial.println("MPU6050 connection failed");
    while (1); // Halt execution if connection fails
  }

  // Set led pin as an output
  pinMode(LED_PIN, OUTPUT);
}

void loop() {
  // Read accelerometer and gyroscope data
  mpu.getMotion6(&ax, &ay, &az, &gx, &gy, &gz);
```

```
// Check if raw values are zero and reinitialize if necessary
if (ax == 0 && ay == 0 && az == 0) {
  // Serial.println("Raw values are zero. Reinitializing MPU6050...");
  mpu.initialize();
} else {
  // Print raw accelerometer data for debugging
  // Serial.print("Raw aX = "); Serial.print(ax);
  // Serial.print(" | Raw aY = "); Serial.print(ay);
  // Serial.print(" | Raw aZ = "); Serial.println(az);

  // Map the raw accelerometer data to 0-255
  data.X = map(ax, -17000, 17000, 0, 255); // X axis data
  data.Y = map(ay, -17000, 17000, 0, 255);
  data.Z = map(az, -17000, 17000, 0, 255); // Z axis data

  // Print mapped accelerometer data
  // Serial.print("Mapped Axis X = ");
  // Serial.print(data.X);
  // Serial.print(" | Mapped Axis Y = ");
  // Serial.print(data.Y);
  // Serial.print(" | Mapped Axis Z = ");
  // Serial.println(data.Z);

  // Check if the value of X is not between 120 and 130
  if (data.X < 80 || data.X > 200) {
  digitalWrite(LED_PIN, HIGH); // Turn on the led
  Serial.println("Accident Alert...");
  } else {
    digitalWrite(LED_PIN, LOW); // Turn off the led
  }

   if (data.Y < 80 || data.Y > 200) {
    digitalWrite(LED_PIN, HIGH); // Turn on the led
    Serial.println("Accident Alert...");
  } else {
    digitalWrite(LED_PIN, LOW); // Turn off the led
  }

  if (data.Z < 150 || data.Z > 350) {
    digitalWrite(LED_PIN, HIGH); // Turn on the led
    Serial.println("Accident Alert...");
```

```
    } else {
      digitalWrite(LED_PIN, LOW); // Turn off the led
    }
  }

  delay(1000); // Delay for readability
}
```

6.1.5 Real life location tracking and Monitoring:

The Qubo Wired Bike GPS Tracker from Hero Group is designed to enhance the safety and security of your bike with advanced features and real-time tracking capabilities. Here are some key features and details about the product:

Real-Time Location Tracking: The tracker allows you to monitor the live location of your bike through the QuboGo app, accessible from anywhere. This feature ensures you always know where your bike is.

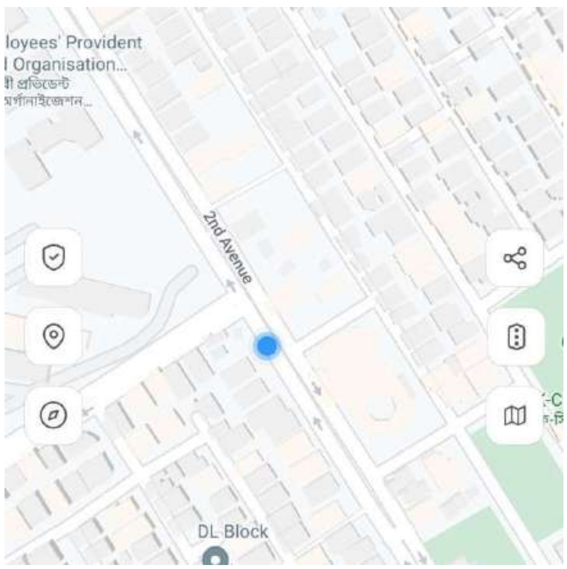


Figure 6.53: Real-Time Location Tracking

Safety Alerts: It includes several alert features such as:

1. **Accident Alerts**: Sends immediate notifications to your emergency contacts if a crash is detected.
2. **Engine On/Off Alerts**: Notifies you whenever your bike's engine is turned on or off, which can help prevent theft.
3. **Tamper Alerts**: Alerts you if there is any attempt to tamper with the GPS device.

Product Introduction:

Qubo GPS tracker in a vehicle is a device that utilizes Global Positioning System (GPS) technology to determine and record the precise location of the vehicle in real-time. This information is then transmitted to a cloud & App, allowing the user to monitor the vehicle's location and

Product Description:

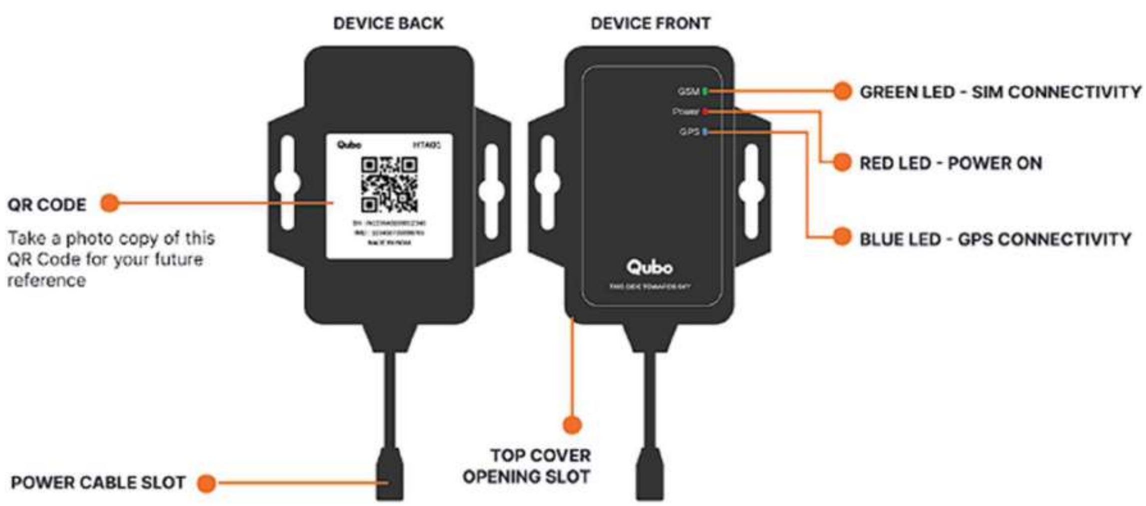


Figure 6.54: Product Description details

Table 6.12 :LED Behavior

Power - Red	Indication	Function
Quick flashing	Flash 0.3s at interval of 0.3s	Low Battery
Solid Red	-	Charging
Flashing	Flash 1s at interval of 3s	Full charge
Slow flashing	Flash 0.1s at interval of 3s	Working Normal
OFF	Quick flashing	Power off

GPS - Blue	Indication	Function
Quick flashing	Flash 0.3s at interval of 0.3s	Searching GPS Signal
Slow flashing	Flash 0.1s at interval of 3s	Receive GPS signal normal
OFF	-	No GPS signal

GSM LED - Green	Indication	Function
Quick flashing	Flash 0.3s at interval of 0.3s	GSM initializing
Flashing	Flash 1s at interval of 3s	Receive GSM signal normal
Solid Green	-	In communication with Phones
Slow flashing	Flash 0.1s at interval of 3s	GSM online
OFF	-	No GSM signal

Installation Process:

Pre-Installation Check

Please make sure that you check the below things before installing the device.

• Do the device installation on your vehicle in an open space for better SIM & GPS connectivity. Avoid doing in the basements.

• Take a Photo of Serial number QR code on the device to use it during the App onboarding.

Installation Steps:

Step 1:

Select a hidden and secure location within the train to mount the GPS tracker. Make sure the top cover of the tracker is facing towards sky and it should be out of sight and not easily accessible to potential thieves. Also, device should be placed in such manner that it should be towards moving direction of Vehicle. [77]

Figure 6.55: The Qubo installment

Step 2

Identify the Positive (+), Ground (-) & ACC wires in the battery. Positive will be the vehicle battery positive (+) terminal, Ground (-) will be the Vehicle metal chassis battery negative (-) terminal, ACC is the vehicle's Ignition ON terminal.

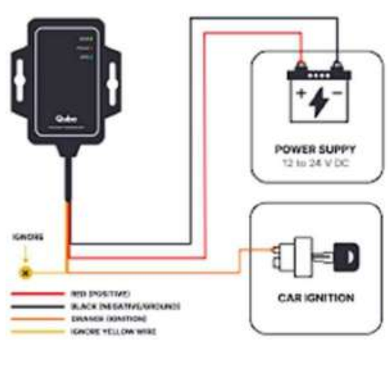


Figure 6.56: The Connection Diagram

Step 3

Connect the Tracker cable to Positive (+), Ground (-) & ACC of train. Red wire to the Battery positive (+) terminal, Black wire to the Ground (-) and orange wire to the Ignition ON/OFF terminal.

Step 4

Connect the other end of Power cable with the Tracker to Power it ON. blinking.

Figure 6.57: tracker and Power cable

We will notice LEDs will start blinking to indicate that device is turned ON. If the LEDs are not blinking, please check the wiring connection again. During power ON, Device will blink for some time & then goes to sleep mode for power saving. In sleep mode, the LEDs wont blink. The device will resume again to normal mode once any activity is detected in the device.

Step 5

You will notice LEDs will start blinking to indicate that device is turned ON. If the LEDs are not blinking, please check the wiring connection again. During power ON, Device will blink for some time & then goes to sleep mode for power saving. In sleep mode, the LEDs wont blink. The device will resume again to normal mode once any activity is detected in the device. Remove the Top cover, switch ON the toggle switch & close the cover back.

Figure 6.58: the toggle switch of GPS

App onboarding:

Download Qubo Go App from App store or Playstore in your mobile.Create an account & add the GPS tracker device. Scan the QR code of device in the App & follow the instructions to commission.QR code will be available on the device & also on the packaging box. Keep a photo of this QR code for your future references.

5. QUBO GO APP:

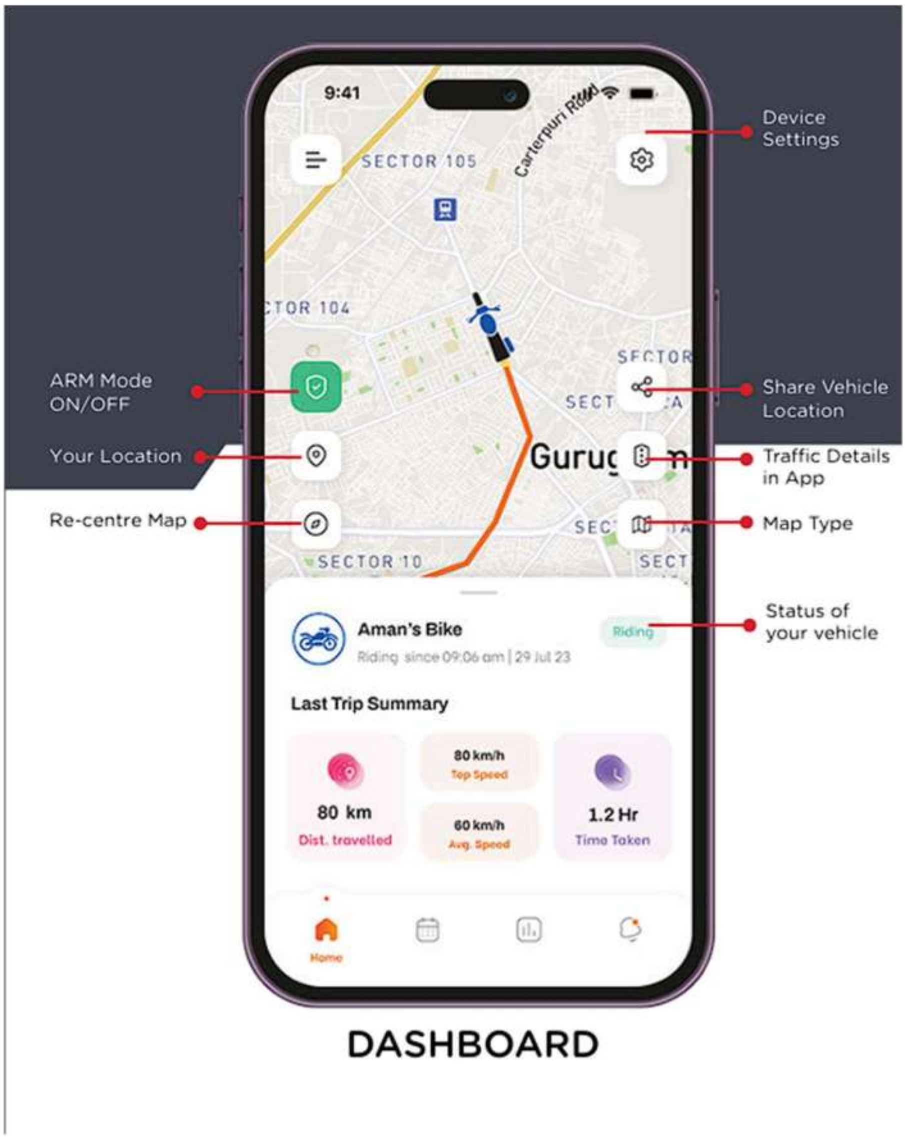


Figure 6.59: The QuboGo app interface

6.2 THE FINAL MODEL

We have used sunboard sheet to make this project and also used a plywood board at base to make it stable. Both the locomotives are same in size and dimension.

Figure 6.60: Design of final model of head on collision prevention

CHATER 7: FUTURE PROSPECT

In our project, we have used two prototype locomotive models for the demonstration purpose of the entire idea that we hold to be suitable for any bigger projects like prevention of head on collision of locomotives. India's railway network is vast and diverse, serving as a critical component of the nation's transportation infrastructure. This automated safety device mainly relies on multiple sensor technology. The sensors **continuously monitor the surrounding environment**. The sensors we used here provide the **real time data about the position speed and trajectory of nearby vehicles**. The data we collected from the sensors helps us to provide a collision prediction algorithm and analyses the vehicle movement patterns and make decision. Our locomotive models have been used as **smaller prototypes of the large locomotives** where head on collision can occur for many causes. Hence the systems that we have used are generally suitable for a small scale. In large scale train for preventing head- on collision we have to alter or improvise some of these methods. Also, the methods should be **efficient, less tedious, least costly, have low maintenance cost** as well as be **power economized.**
We shall thus go step by step into each of the systems that we have used in our project and see what are the modifications that can be done when this is implemented on a large scale. In total, we have four systems namely:

1. **Locomotive Sensing and Automatic Braking System**
2. **Accident Prevention Signal**
3. **Accident Alert System**
4. **Real-Time Location Tracking and Monitoring**

Locomotive Sensing and Automatic Braking System:

Sensors:

In our head- on collision prevention model we used a variety of sensors for different locomotive and obstacle sensing. In real locomotive we can install different sensors to make it more efficient. As locomotive has different sensors init from the beginning we can use those to make it less costly. As locomotive haveto go through different harsh weather, mechanical stress, electrical interference,data managements, calibration problem. So most efficient sensors will be used here.

Obstacle Detection Sensors:

5. **Lidar (Light Detection and Ranging)**: Lidar sensors use laser beams to detect obstacles on the tracks by measuring the time it takes for the laser to bounce back from an object.
6. **Radar**: Radar sensors use radio waves to detect the presence, distance, and speed of objects on the tracks.
7. **Ultrasonic Sensors**: These sensors use ultrasonic waves to detect objects close to the train, providing data on potential obstacles.
8. **AR3000**: The AR 3000 can be installed at the front and rear of trains to detect obstacles on the tracks, providing critical data for collision avoidance systems.

Track Condition Sensors:

- **Infrared Sensors**: Infrared sensors can detect heat signatures from objectsor obstructions on the tracks.
- **Cameras**: Vision systems using high-resolution cameras can visually detect obstacles and provide real-time images for further processing.

Train Condition Monitoring Sensors:

9. **Accelerometers**: Accelerometers measure the train's acceleration and can detect sudden changes indicative of an impact or derailment.
10. **Gyroscopes**: Gyroscopes measure the train's orientation and can detect tilting or rolling that could indicate a derailment.
11. **Pressure Sensors**: These sensors monitor air pressure in the braking system to detect anomalies that might require an emergency stop.

Environmental Sensors:

12. **Weather Sensors**: These sensors monitor weather conditions (rain, snow, fog) that could affect train safety and trigger an emergency braking response if conditions are deemed hazardous.
13. **Vibration Sensors**: Vibration sensors can detect unusual vibrations that might indicate track or train issues requiring an emergency stop.

Automatic Braking:

Types of Automatic Braking Systems:

- **Automatic Train Protection (ATP)**: This system ensures that trains operate within safe limits by monitoring speed, signals, and otherparameters. It can automatically apply brakes if a train exceeds the speed limit or fails to respond to signal indications.
- **Train Collision Avoidance System (TCAS)**: This system helps prevent train collisions by automatically applying brakes if it detects another train on the same track within a certain distance.

- **Emergency Brake Application (EBA)**: This is an automatic response system that applies brakes when an emergency situation is detected, such asa sudden obstruction on the tracks or a critical system failure.

Automatic Train Control (ATC) systems, apply brakes to the wheels through a series of coordinated actions involving sensors, control units, and braking mechanisms. Here's a detailed flowchart of how this process works:

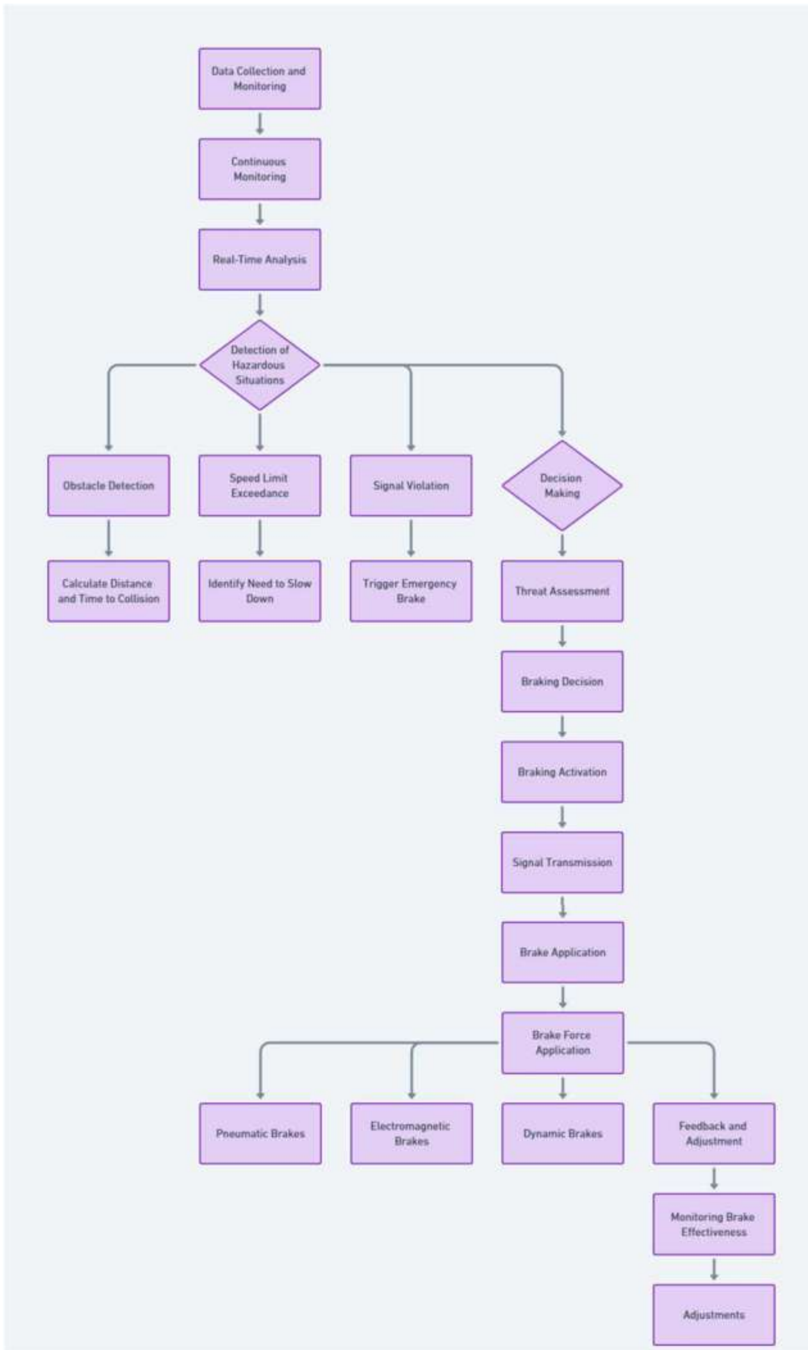


Figure 7.1: The flowchart of automatic braking

Case1: When both locomotive are in the same speed

Given Data:
- Initial distance between the two trains: = 3 km= 3000 m
- Speed of each train: V = 100 km/h = (100 * 1000)/3600 = 27.78 m/s
- Desired final distance between the two trains: = 500 m

Step-by-Step Solution:

1. Calculate the total distance each train needs to cover while braking:

Since the final distance between the trains should be 500 meters, the total distance both trains need to cover is:

Total covered=initial- final = 3000m- 500m= 2500m

Each train will cover half of this distance while braking:
covered per train = 2500 m\ 2 = 1250 \ m

2.Calculate the deceleration rate required for each train:

Using the kinematic equation for uniform deceleration:

V^2 = U^2 + 2aD

where V is the final velocity (0 m/s), U is the initial velocity (27.78 m/s), a is the deceleration, and D is the distance (750 m). Rearranging for a

0 = (27.78)^2 + 2 * a * 1250
a = -0.30869 m/s^2

3.Calculate the time required to stop:

Using the equation:

V = U + at

Rearranging for t:

$0 = 27.78 + (-0.30869) * t$

$t = 89.99$ seconds

Case2: When both locomotive are in the Different speed

Given Data:

Initial distance between the two trains: $D_{initial} = 3$ km $= 3000$ m

Speed of train1: $V1 = 100$ km/h $= (100 * 1000)/3600 = 27.78$ m/s

Speed of train 2 : $V2 = 80$ km/h$= (80*1000)/3600 = 22.22$ m/s

Desired final distance between the two trains: final=500m

Step-by-Step Solution:

Calculate the total distance each train needs to cover while braking:

Since the final distance between the trains should be 500 meters, the total distance both trains need to cover is:

$D_{total\ covered} = D_{initial} - D_{final} = 3000\ m - 500\ m = 2500$

Calculate the braking distance for each train:

Let D1andD2 be the distances covered by train 1 and train 2 respectively.

Since both trains are braking to maintain a combined stopping distance of 2500 meters, and the ratio of their initial speeds determines how far each one travels:

$D1/D2 = V1/V2$

Substituting the speeds:

$D1/D2 = 27.78/22.22 = 5/4$

Therefore:

$D1=(5/4)*D2$

And:

$D1+D2=2500$

Substituting D1:

$(5/4)D2+D2=2500$

$(9/4)*D2=2500$

$D2=(2500\times4)/9=1111.11$

$D1=(5/4)\times1111.11=1388.89$

Calculate the deceleration required for each train:

Using the kinematic equation for uniform deceleration:

$V^2=U^2+2aD$

For train 1:

$0=(27.78)^2+2\cdot a1\cdot 1388.890$

$a1=-0.278\ m/s^2$

For train 2:

$0=(22.22)2+2\cdot a2\cdot 1111.110$

$a2=-0.222\ m/s^2$

Calculate the time required to stop:

Using the equation:

$V=U+at$

For train 1:

$0=27.78+(-0.278)\cdot t$

$t=100$ seconds

For train 2:

$0=22.22+(-0.222)\cdot t$

$t=100$ seconds

Accident Prevention Signal:

In our locomotive prototype we have used two ultrasonic sensors for accident prevention signal by speed sensing. Our concept incorporates two ultrasonic sensors in the track together with an accident prevention signal. This particular model is intended to identify the existence of another locomotive within the same track. Using an Arduino, it can also gauge how quickly a locomotive moves between two ultrasonic sensors. The system consists of an LED indication and two ultrasonic sensors. In the event that another train approaches on the same track, the LED will alert the locomotive.

In real world we can use different sensors to measure the speed and transmit the data to opposite locomotive for accident alert.

Sensors:

Lidar: Light detection and ranging sensors can measure distances and detect obstacles or obstructions on the tracks.

Infrared Cameras: Infrared cameras can detect heat signatures, useful for detecting potential hazards such as overheated components or track anomalies.

Ultrasonic Sensors: Ultrasonic sensors can detect the presence of objects or obstacles by emitting high-frequency sound waves and analyzing their reflections.

Communication Systems:

Wireless Transmitters/Receivers: These systems enable communication between locomotives, trackside equipment, and centralized control centers, facilitating the exchange of safety-critical information.

Satellite Communication: Satellite communication can provide reliable connectivity in remote areas where traditional communication infrastructuremay be lacking or unreliable.

Onboard Computers and Processors:

High-performance onboard computers and processors are essential for processing data from various sensors, running AI algorithms for hazard detection, and making real-time decisions to prevent accidents.

Display Interfaces:

Visual display interfaces, such as screens or heads-up displays, can provide train operators with real-time information about potential hazards detected by the accident alert system.

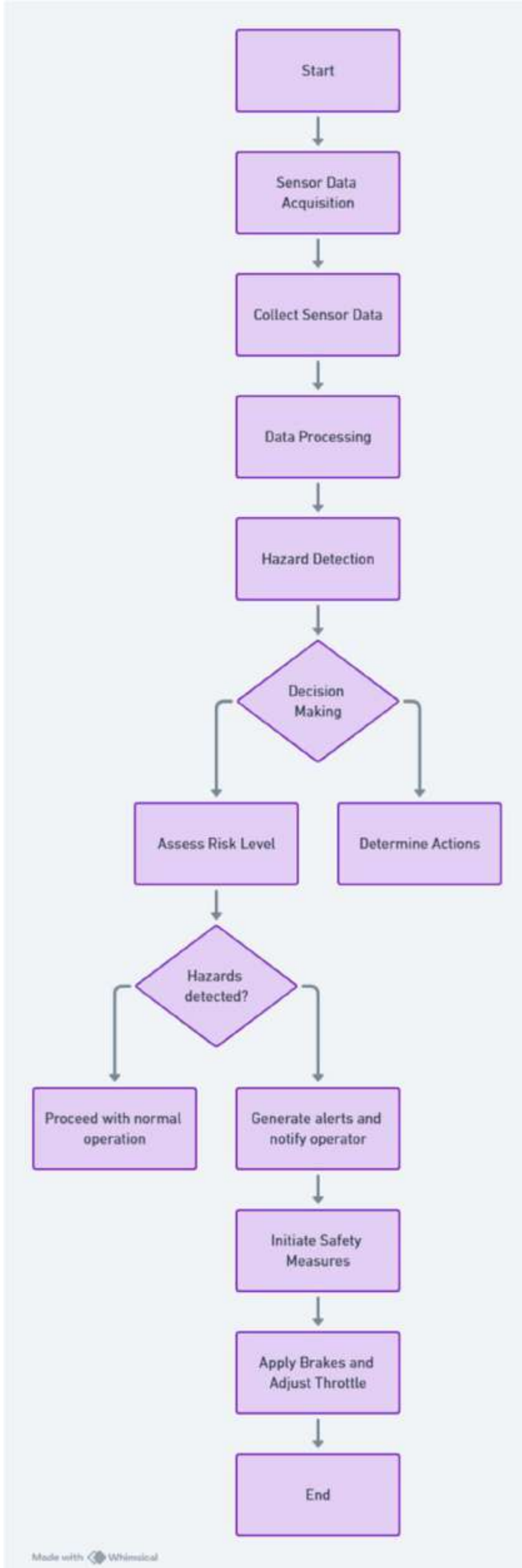


Figure 7.2: The flowchart of Automatic prevention signal

Accident alert system

In this system, a gyroscope sensor will be mounted on the locomotive to continuously monitor its orientation and movement. The data will be collected by the sensor is processed and analyzed against predefined safety thresholds. When the sensor detects abnormal or hazardous movement indicative of a potential collision, it triggers a wireless signal transmission to the control room. The control room, upon receiving the alert, can take necessary preventive actions such as activating the emergency braking system and notifying nearby locomotives.

This innovative approach not only enhances the situational awareness of locomotive operators but also significantly reduces the response time in emergency situations.

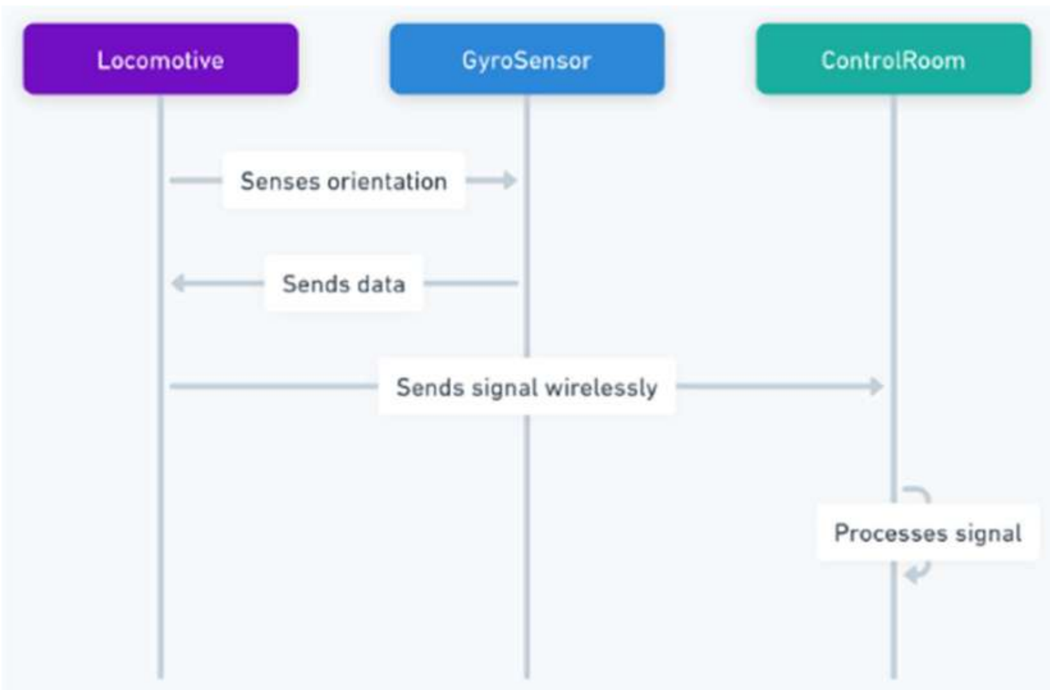


Figure 7.3: The connection between Locomotive and Control Room

- Gyroscope sensors continuously monitor the orientation and angular velocity of locomotives in real-time.
- Data from the sensors are transmitted wirelessly to the control room interface.
- In the control room, the received data is processed and analyzed to detect any anomalies suggestive of a head-on collision risk.
- If a potential collision scenario is identified, an alert is promptly generated, notifying operators in the control room.

- Operators can take immediate preventive actions, such as issuing warnings to locomotive operators, adjusting speeds, or initiating emergency braking procedures to avoid the collision.

Flowchart:

we can use a flowchart to model the process of detecting and alerting a potential head-on collision using a gyroscope sensor. The flowchart will include the following steps:

- **Gyroscope Sensor Activation**
- **Data Processing Unit (DPU)**
- **Threshold Analysis**
- **Wireless Signal Transmission**
- **Control Room Alert**
- **Preventive Action**

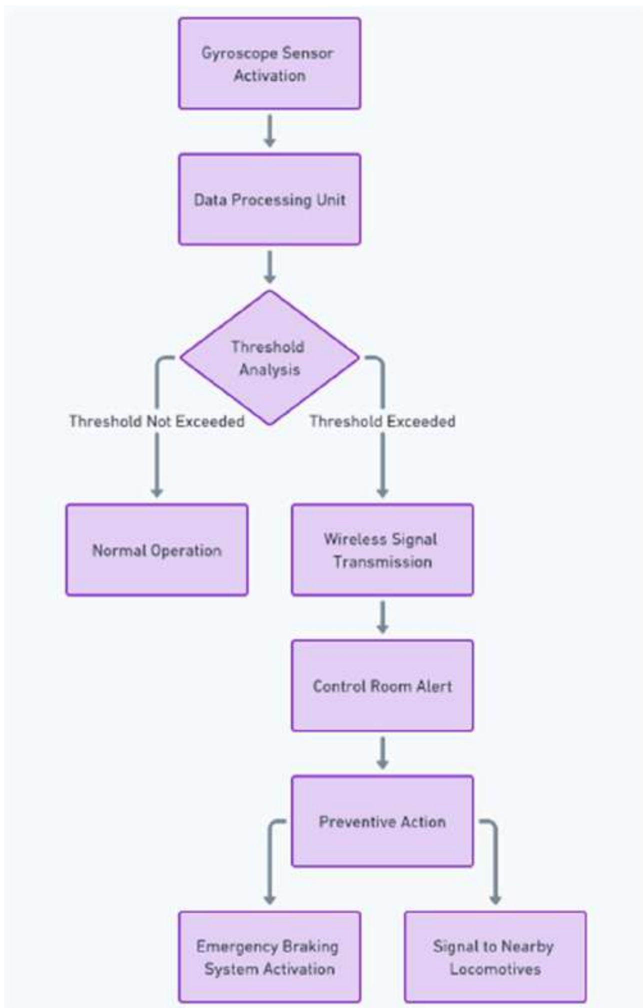


Figure 7.4: flowchart of Accident alert System

Benefits:

11. Enhanced Safety: By detecting and preventing head-on collisions, the Accident Alert System significantly enhances the safety of railwayoperations, reducing the risk of injuries, fatalities, and infrastructure damage.
12. Operational Efficiency: Swift detection and response to potential collision risks optimize operational efficiency, minimizing disruptions to railway services and ensuring timely transportation of goods and passengers.
13. Cost Savings: Preventing accidents through proactive measures not only saves lives but also mitigates the financial costs associated with accident investigations, legal liabilities, and infrastructure repairs.

The proposed Accident Alert System, leveraging gyroscope sensors and wireless communication technology, represents a crucial advancement in head- on collision prevention for locomotives. By providing early detection and prompt alerts to operators, this system plays a pivotal role in ensuring the safety and efficiency of railway transportation networks.

Real time location tracking and monitoring:

The future of real-time location tracking and monitoring promises significant advancements driven by technological innovations and evolving societal needs. Real-time location tracking and monitoring in locomotives is set to revolutionize the rail industry, enhancing safety, efficiency, and operational effectiveness. Here we have used QuboGo GPS for tracking the locomotive. In real locomotive different gps and cellular module will be used for tracking it.

- ### Global Navigation Satellite System (GNSS) Receivers

14. **High-Precision GNSS Receivers**: Provide accurate location data using multiple satellite constellations (GPS, GLONASS, Galileo, etc.).
15. **Differential GNSS (DGNSS) Systems**: Enhance accuracy through ground-based reference stations that correct satellite signals.

2. IoT Sensors and Devices

- **Environmental Sensors**: Monitor temperature, humidity, and other environmental parameters within the locomotive and along the tracks.
- **Vibration and Shock Sensors**: Detect vibrations and shocks to identify potential mechanical issues.
- **Position and Speed Sensors**: Provide data on the train's position and speed in real time.

3. 5G and Cellular Modems

- **5G Modems**: Enable high-speed, low-latency communication essential for real-time data transmission.
- **Multi-Network Cellular Modems**: Ensure connectivity by switching between available cellular networks.

4. Communication Systems

- **Radio Communication Systems**: Maintain communication between the train and control centers.
- **Automatic Train Control (ATC) Systems**: Use real-time data to manage train speeds and ensure safe operations.

5. Rail Traffic Management Systems (RTMS)

- **Centralized Control Systems**: Monitor and manage train movements using real-time location data.
- **Dynamic Scheduling Tools**: Adjust train schedules based on real-time data to optimize operations.

6. Collision Avoidance and Safety Systems

- **Positive Train Control (PTC)**: Automatically control train movements to prevent collisions and derailments.
- **Obstacle Detection Systems**: Use radar, lidar, and cameras to detect obstacles on the tracks.

7. High-Resolution Mapping Tools

- **3D Mapping Software**: Provide detailed maps for precise navigation and tracking.
- **Geospatial Information Systems (GIS)**: Integrate location data with geographic information for comprehensive analysis.

By leveraging this combination of equipment and technologies, the rail industry can achieve robust real-time location tracking and monitoring, leading to safer, more efficient, and reliable train operations.

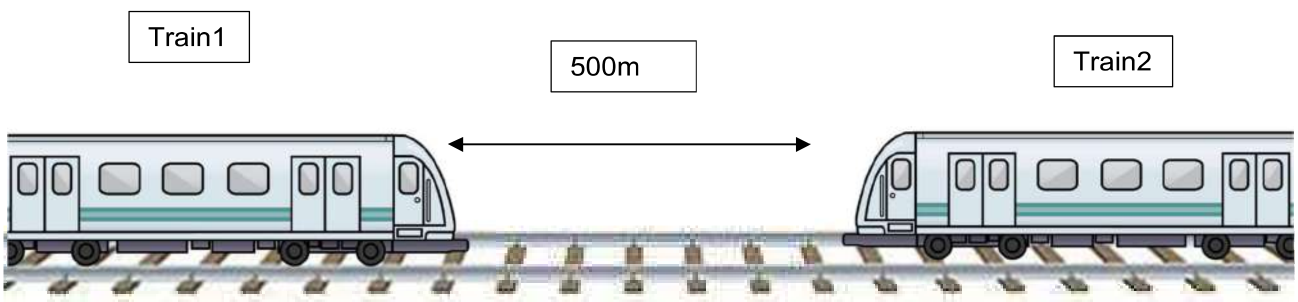


Figure: 7.5: Train collision avoidance model

CHAPTER 8: CONCLUSION

Our project "**THE HEAD ON COLLISION PREVENTION MODEL (LOCOMOTIVES)**" is currently in the developing stages. It now has the ability to detect any other locomotives in the same track coming from any side, to alert if any accident occurs and to prevent the accident by sensing the speed and locationof any train. As mentioned in the previous chapter **"FUTURE PROSPECTS",** we have built our project with the help of locomotive models made by sunboard sheet. These represent the miniature prototypes of the actual locomotives that are already present or will be developed in the near future. For more numbers of locomotives where this project can be used for preventing many accidents and saving thousands of life.

Our project is low- cost model which will provide the maximum safety and fast service than the previous models in very less amount of money. Not only that it will be a multi-service model which will provide everything.

Further the communication system has to be developed more so as to ensure a more reliable operation and effortless prevention of the head on collision. Further as we had previously mentioned that real time location tracking and automatic braking that need to be developed further.

This automatic braking will be of great help when a good yield is the main concern. However, in the beginning stages, our miniature model will serve as an example and our technologies and procedures described in **"FUTURE PROSPECTS"** will help in establishing a head-on collision prevention model which can easily produce a good yield with some supervision.

To end, we hope that our project has the capability to transform the current scenario to prevent head on collision in locomotives in India and outside and with further developments it will serve as a major development in this field which can saves not only thousands of life but it also save extra the financial expenses.

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